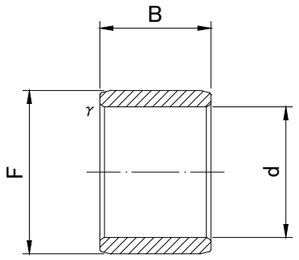
INNER RINGS

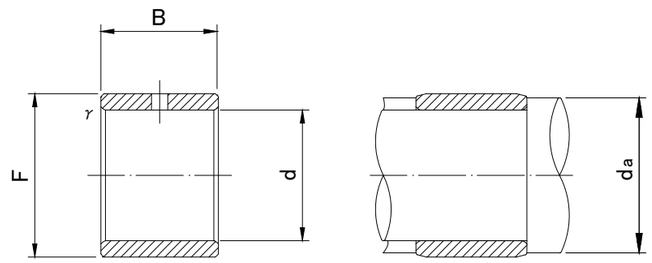






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IR,IRZ TYPE

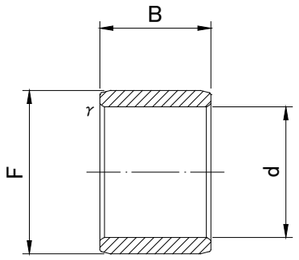
Shaft Diameter (mm)	Designation		Dimensions (mm)				Standard mounting dimensions (mm)		Mass	Usable bearing designation					
	IR	IRZ	d	F	B	*r*s min	da MIN	da MAX	g (approx)	RNA 49	RNA 59	RNA 69	RNA 48	NK	RNAF
17	IR 172016.5	—	17	20	16.5	0.3	19	19	11.1	—	—	—	—	—	—
	IR 172020	—	17	20	20	0.3	19	19	14	—	—	—	—	—	—
	IR 172020.5	—	17	20	20.5	0.3	19	19	13.7	—	—	—	—	—	—
	IR 172030.5	—	17	20	30.5	0.3	19	19	20.5	—	—	—	—	—	—
	IR 172116	—	17	21	16	0.3	19	20	14.5	—	—	—	—	NK 21/16	—
	IR 172120	—	17	21	20	0.3	19	20	18	—	—	—	—	NK 21/20	—
	IR 172213	—	17	22	13	0.3	19	21	15.5	RNA 4903	—	—	—	—	RNAF 223013
	—	IRZ 172214	17	22	14	0.3	19	21	16.5	RNA 4903UU	—	—	—	—	—
	IR 172215.5	—	17 0 -0.008	22	15.5	0.3	19	21	18.3	—	—	—	—	—	—
	IR 172216	—	17	22	16	0.3	19	21	19	—	—	—	—	—	RNAF 223516
	IR 172216.5	—	17	22	16.5	0.3	19	21	19.4	—	—	—	—	—	—
	IR 172218	—	17	22	18	0.3	19	21	21	—	RNA 5903	—	—	—	—
	IR 172223	—	17	22	23	0.3	19	21	26.5	—	—	RNA 6903	—	—	—
	—	IRZ 172224	17	22	24	0.3	19	21	28	—	—	RNA 6903UU	—	—	—
	IR 172225.5	—	17	22	25.5	0.3	19	21	30	—	—	—	—	—	—
	IR 172226	—	17	22	26	0.3	19	21	31	—	—	—	—	—	RNAFW 223026
	IR 172232	—	17	22	32	0.3	19	21	38	—	—	—	—	—	RNAFW 223532
20	IR 202416	—	20	24	16	0.3	22	23	16.5	—	—	—	—	NK 24/16	—
	IR 202416.5	—	20	24	16.5	0.3	22	23	17.5	—	—	—	—	—	—
	IR 202420	—	20	24	20	0.3	22	23	20.5	—	—	—	—	NK 24/20	—
	IR 202420.5	—	20	24	20.5	0.3	22	23	22	—	—	—	—	—	—
	IR 202510.5	—	20	25	10.5	0.3	22	24	14.3	—	—	—	—	—	—
	IR 202515.5	—	20	25	15.5	0.3	22	24	21	—	—	—	—	—	—
	IR 202516	—	20	25	16	0.3	22	24	22	—	—	—	—	—	RNAF 253716
	IR 202517	—	20	25	17	0.3	22	24	23	RNA 4904	—	—	—	—	RNAF 253517
	—	IRZ 202518	20 0	25	18	0.3	22	24	24	RNA 4904UU	—	—	—	—	—
	IR 202520	—	20 -0.010	25	20	0.3	22	24	27	—	—	—	—	—	—
	IR 202520.5	—	20	25	20.5	0.3	22	24	28	—	—	—	—	—	—
	IR 202523	—	20	25	23	0.3	22	24	31	—	RNA 5904	—	—	—	—
	IR 202525.5	—	20	25	25.5	0.3	22	24	34.5	—	—	—	—	—	—
	IR 202526	—	20	25	26	0.3	22	24	36	—	—	—	—	—	RNAFW 253526
	IR 202526.5	—	20	25	26.5	0.3	22	24	36	—	—	—	—	—	—
	IR 202530	—	20	25	30	0.3	22	24	40.5	—	—	RNA 6904	—	—	—
	—	IRZ 202531	20	25	31	0.3	22	24	41.5	—	—	RNA 6904UU	—	—	—
	IR 202532	—	20	25	32	0.3	22	24	44	—	—	—	—	—	RNAFW 253732

IR,IRZ

INNER RINGS

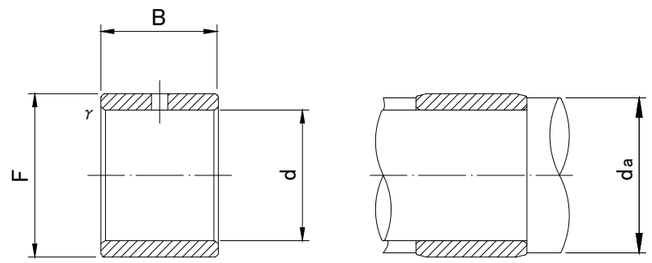






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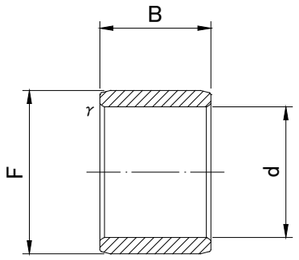
IR,IRZ TYPE

Shaft Diameter (mm)	Designation		Dimensions (mm)				Standard mounting dimensions (mm)		Mass	Usable bearing designation					
	IR	IRZ	d	F	B	*r*s min	da MIN	da MAX	g (approx)	RNA 49	RNA 59	RNA 69	RNA 48	NK	RNAF
22	**IR 222616**	—	22	26	16	0.3	24	25	17.5	—	—	—	—	NK 26/16	—
	IR 222620	—	22	26	20	0.3	24	25	24	—	—	—	—	NK 26/20	—
	IR 222817	—	22 0	28	17	0.3	24	27	30.5	RNA 49/22	—	—	—	—	—
	IR 222820.5	—	22 -0.010	28	20.5	0.3	24	27	37	—	—	—	—	—	—
	IR 222823	—	22	28	23	0.3	24	27	42	—	RNA 59/22	—	—	—	—
	IR 222830	—	22	28	30	0.3	24	27	55	—	—	RNA 69/22	—	—	—
25	**IR 252920**	—	25	29	20	0.3	27	28	25	—	—	—	—	NK 29/20	—
	IR 252930	—	25	29	30	0.3	27	28	38	—	—	—	—	NK 29/30	—
	IR 253015	—	25	30	15	0.3	24	29	24.5	—	—	—	—	—	—
	IR 253015.5	—	25	30	15.5	0.3	24	29	25.5	—	—	—	—	—	—
	IR 253016	—	25	30	16	0.3	27	29	28	—	—	—	—	—	RNAF 304216
	IR 253017	—	25	30	17	0.3	27	29	28.5	RNA 4905	—	—	—	—	RNAF 304017
	—	**IRZ 253018**	25	30	18	0.3	27	29	29.5	RNA 4905UU	—	—	—	—	—
	IR 253020	—	25	30	20	0.3	24	29	33	—	—	—	—	—	—
	IR 253020.5	—	25 0	30	20.5	0.3	24	29	34	—	—	—	—	—	—
	IR 253023	—	25 -0.010	30	23	0.3	27	29	38	—	RNA 5905	—	—	—	—
	IR 253025.5	—	25	30	25.5	0.3	24	29	42.5	—	—	—	—	—	—
	IR 253026	—	25	30	26	0.3	27	29	44.5	—	—	—	—	—	RNAFW 304026
	IR 253026.5	—	25	30	26.5	0.3	24	29	44	—	—	—	—	—	—
	IR 253030	—	25	30	30	0.3	27	29	49	—	—	RNA 6905	—	—	—
	IR 253030.5	—	25	30	30.5	0.3	24	29	50.5	—	—	—	—	—	—
	—	**IRZ 253031**	25	30	31	0.3	27	29	51	—	—	RNA 6905UU	—	—	—
	IR 253032	—	25	30	32	0.3	27	29	54	—	—	—	—	—	RNAFW 304232
	IR 253038.5	—	25	30	38.5	0.3	24	29	64	—	—	—	—	—	—
28	**IR 283217**	—	28	32	17	0.3	30	31	24.5	RNA 49/28	—	—	—	—	—
	IR 283220	—	28	32	20	0.3	30	31	28.5	—	—	—	—	NK 32/20	—
	IR 283220.5	—	28 0	32	20.5	0.3	30	31	29.5	—	—	—	—	—	—
	IR 283223	—	28 -0.010	32	23	0.3	30	31	34	—	RNA 59/28	—	—	—	—
	IR 283230	—	28	32	30	0.3	30	31	43	—	—	RNA 69/28	—	NK 32/30	—
	IR 283230.5	—	28	32	30.5	0.3	30	31	44	—	—	—	—	—	—

INNER RINGS

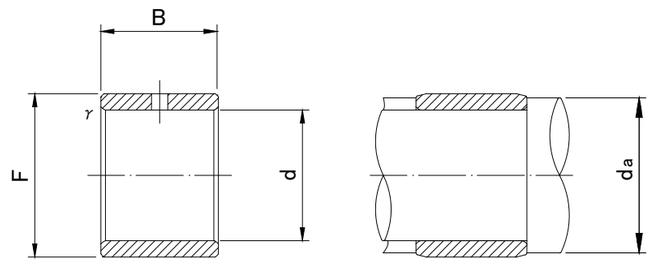






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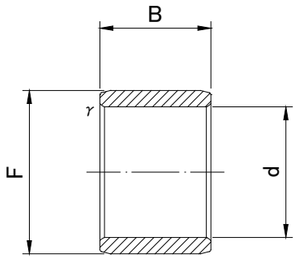
IR,IRZ TYPE

Shaft Diameter (mm)	Designation		Dimensions (mm)				Standard mounting dimensions (mm)		Mass	Usable bearing designation					
	IR	IRZ	d	F	B	r s min	da MIN	da MAX	g (approx)	RNA 49	RNA 59	RNA 69	RNA 48	NK	RNAF
30	IR 303516	—	30	35	16	0.3	32	34	31.5	—	—	—	—	—	RNAF 354716
	IR 303517	—	30	35	17	0.3	32	34	33.5	RNA 4906	—	—	—	—	RNAF 354517
	—	IRZ 303518	30	35	18	0.3	32	34	35	RNA 4906UU	—	—	—	—	—
	IR 303520	—	30	35	20	0.3	32	34	38.5	—	—	—	—	NK 35/20	—
	IR 303520.5	—	30	35	20.5	0.3	34	34	40	—	—	—	—	—	—
	IR 303523	—	30	35	23	0.3	32	34	44	—	RNA 5906	—	—	—	—
	IR 303525.5	—	30 0 -0.010	35	25.5	0.3	34	34	51	—	—	—	—	—	—
	IR 303526	—	30	35	26	0.3	32	34	52	—	—	—	—	—	RNAFW 354526
	IR 303530	—	30	35	30	0.3	32	34	59	—	—	RNA 6906	—	NK 35/30	—
	IR 303530.5	—	30	35	30.5	0.3	34	34	60	—	—	—	—	—	—
	—	IRZ 303531	30	35	31	0.3	32	34	61	—	—	RNA 6906UU	—	—	—
	IR 303532	—	30	35	32	0.3	32	34	64	—	—	—	—	—	RNAFW 354732
	IR 303820	—	30	38	20	0.6	34	37	65	—	—	—	—	—	—
32	IR 323720	—	32	37	20	0.3	34	36	43.5	—	—	—	—	NK 37/20	—
	IR 323730	—	32	37	30	0.3	34	36	63	—	—	—	—	NK 37/30	—
	IR 324020	—	32 0 -0.012	40	20	0.6	36	39	69	RNA 49/32	—	—	—	—	—
	IR 324027	—	32	40	27	0.6	36	39	92	—	RNA 59/32	—	—	—	—
	IR 324036	—	32	40	36	0.6	36	39	123	—	—	RNA 69/32	—	—	—
35	IR 354017	—	35	40	17	0.3	37	39	39	—	—	—	—	—	RNAF 405017
	IR 354020	—	35	40	20	0.3	37	39	46	—	—	—	—	NK 40/20	RNAF 405520
	IR 354020.5	—	35	40	20.5	0.3	39	39	46.5	—	—	—	—	—	—
	IR 354025.5	—	35	40	25.5	0.3	39	39	46	—	—	—	—	—	—
	IR 354030	—	35	40	30	0.3	37	39	67	—	—	—	—	NK 40/30	—
	IR 354034	—	35 0	40	34	0.3	37	39	78	—	—	—	—	—	RNAFW 405034
	IR 354040	—	35 -0.012	40	40	0.3	37	39	95	—	—	—	—	—	RNAFW 405540
	IR 354220	—	35	42	20	0.6	39	41	65	RNA 4907	—	—	—	—	—
	—	IRZ 354221	35	42	21	0.6	39	41	67	RNA 4907UU	—	—	—	—	—
	IR 354227	—	35	42	27	0.6	39	41	80	—	RNA 5907	—	—	—	—
	IR 354236	—	35	42	36	0.6	39	41	120	—	—	RNA 6907	—	—	—
	—	IRZ 354237	35	42	37	0.6	39	41	120	—	—	RNA 6907UU	—	—	—
38	IR 384320	—	38 0	43	20	0.3	40	42	49.5	—	—	—	—	NK 43/20	—
	IR 384330	—	38 -0.012	43	30	0.3	40	42	72	—	—	—	—	NK 43/30	—

INNER RINGS

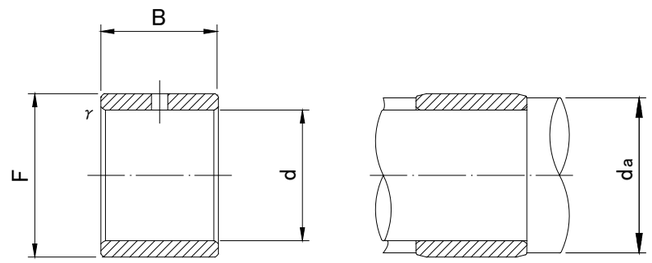






IR





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IR,IRZ TYPE

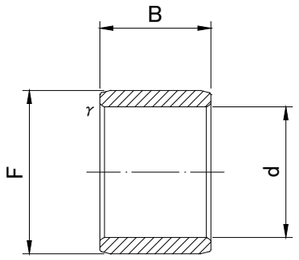
Shaft Diameter (mm)	Designation		Dimensions (mm)				Standard mounting dimensions (mm)		Mass	Usable bearing designation					
	IR	IRZ	d	F	B	r s min	da MIN	da MAX	g (approx)	RNA 49	RNA 59	RNA 69	RNA 48	NK	RNAF
40	IR 404517	—	40	45	17	0.3	42	44	44.5	—	—	—	—	—	RNAF 455517
	IR 404520	—	40	45	20	0.3	42	44	51	—	—	—	—	NK 45/20	RNAF 456220
	IR 404520.5	—	40	45	20.5	0.6	44	44	52.5	—	—	—	—	—	—
	IR 404525.5	—	40	45	25.5	0.6	44	44	65.5	—	—	—	—	—	—
	IR 404530	—	40	45	30	0.3	42	44	77	—	—	—	—	NK 45/30	—
	IR 404530.5	—	40	45	30.5	0.6	44	44	78.5	—	—	—	—	—	—
	IR 404534	—	40 0 -0.012	45	34	0.3	42	44	88	—	—	—	—	—	RNAFW 455534
	IR 404540	—	40	45	40	0.3	42	44	105	—	—	—	—	—	RNAFW 456240
	IR 404540.5	—	40	45	40.5	0.6	44	44	104	—	—	—	—	—	—
	IR 404822	—	40	48	22	0.6	44	47	93	RNA 4908	—	—	—	—	—
	—	IRZ 404823	40	48	23	0.6	44	47	95	RNA 4908UU	—	—	—	—	—
	IR 404830	—	40	48	30	0.6	44	47	123	—	RNA 5908	—	—	—	—
	IR 404840	—	40	48	40	0.6	44	47	165	—	—	RNA 6908	—	—	—
	—	IRZ 404841	40	48	41	0.6	44	47	170	—	—	RNA 6908UU	—	—	—
42	IR 424720	—	42 0 -0.012	47	20	0.3	44	46	54	—	—	—	—	NK 47/20	—
	IR 424730	—	42	47	30	0.3	44	46	81	—	—	—	—	NK 47/30	—
45	IR 455020	—	45	50	20	0.3	47	49	58	—	—	—	—	—	RNAF 506220
	IR 455025	—	45	50	25	0.6	49	49.5	71	—	—	—	—	NK 50/25	—
	IR 455025.5	—	45	50	25.5	0.6	44	49	73	—	—	—	—	—	—
	IR 455030.5	—	45	50	30.5	0.6	44	49	87.5	—	—	—	—	—	—
	IR 455035	—	45	50	35	0.6	49	49.5	95	—	—	—	—	NK 50/35	—
	IR 455040	—	45	50	40	0.3	49	49.5	115	—	—	—	—	—	RNAFW 506240
	IR 455222	—	45 0 -0.012	52	22	0.6	49	51	88	RNA 4909	—	—	—	—	—
	—	IRZ 455223	45	52	23	0.6	49	51	93	RNA 4909UU	—	—	—	—	—
	IR 455230	—	45	52	30	0.6	49	51	123	—	RNA 5909	—	—	—	—
	IR 455240	—	45	52	40	0.6	49	51	165	—	—	RNA 6909	—	—	—
	—	IRZ 455241	45	52	41	0.6	49	51	170	—	—	RNA 6909UU	—	—	—
	IR 455520	—	45	55	20	0.6	49	54	120	—	—	—	—	—	RNAF 557220
	IR 455522	—	45	55	22	1	50	54	130	—	—	—	—	—	
	IR 455540	—	45	55	40	0.6	49	54	245	—	—	—	—	—	RNAFW 557240

IR,IRZ

INNER RINGS

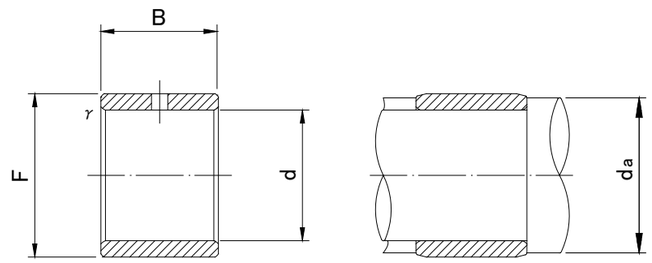






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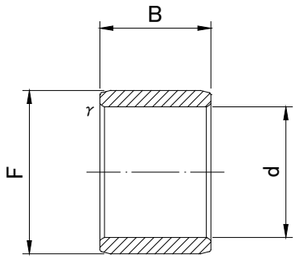
IR,IRZ TYPE

Shaft Diameter (mm)	Designation		Dimensions (mm)				Standard mounting dimensions (mm)		Mass	Usable bearing designation					
	IR	IRZ	d	F	B	*r*s min	da MIN	da MAX	g (approx)	RNA 49	RNA 59	RNA 69	RNA 48	NK	RNAF
50	**IR 505520**	—	50	55	20	0.6	54	54.5	63	—	—	—	—	—	RNAF 556820
	IR 505525	—		55	25	0.6	54	54.5	77	—	—	—	—	NK 55/25	—
	IR 505535	—		55	35	0.6	54	54.5	110	—	—	—	—	NK 55/35	—
	IR 505540	—		55	40	0.6	54	54.5	130	—	—	—	—	—	RNAFW 556840
	IR 505822	—		58	22	0.6	54	57	116	RNA 4910	—	—	—	—	—
	—	**IRZ 505823**	0	58	23	0.6	54	57	118	RNA 4910UU	—	—	—	—	—
	IR 505830	—	-0.012	58	30	0.6	54	57	159	—	RNA 5910	—	—	—	—
	IR 505840	—		58	40	0.6	54	57	210	—	—	RNA 6910	—	—	—
	—	**IRZ 505841**		58	41	0.6	54	57	215	—	—	RNA 6910UU	—	—	—
	IR 506020	—		60	20	1	55	59	135	—	—	—	—	—	RNAF 607820
	IR 506025	—		60	25	1	55	59	163	—	—	—	—	—	—
	IR 506040	—		60	40	1	55	59	265	—	—	—	—	—	RNAFW 607840
55	**IR 556025**	—	55	60	25	0.6	59	59.5	88	—	—	—	—	NK 60/25	—
	IR 556035	—		60	35	0.6	59	59.5	120	—	—	—	—	NK 60/35	—
	IR 556325	—		63	25	1	60	61	145	RNA 4911	—	—	—	—	—
	IR 556334	—	0	63	34	1	60	61	192	—	RNA 5911	—	—	—	—
	IR 556345	—	-0.015	63	45	1	60	61	255	—	—	RNA 6911	—	—	—
	IR 556530	—		65	30	1	60	63	220	—	—	—	—	—	RNAF 658530
	IR 556560	—		65	60	1	60	63	435	—	—	—	—	—	RNAFW 658560
60	**IR 606825**	—	60	68	25	0.6	64	66	150	RNA 4912	—	—	—	NK 68/25	—
	IR 606834	—		68	34	1	65	66	206	—	RNA 5912	—	—	—	—
	IR 606835	—		68	35	0.6	64	66	210	—	—	—	—	NK 68/35	—
	IR 606845	—	0	68	45	1	65	66	275	—	—	RNA 6912	—	—	—
	IR 607025	—	-0.015	70	25	1	65	68	195	—	—	—	—	—	—
	IR 607030	—		70	30	1	65	68	240	—	—	—	—	—	RNAF 709030
	IR 607060	—		70	60	1	65	68	480	—	—	—	—	—	RNAFW 709060
65	**IR 657225**	—	65	72	25	1	70	70.5	145	RNA 4913	—	—	—	—	—
	IR 657234	—		72	34	1	70	70.5	193	—	RNA 5913	—	—	—	—
	IR 657245	—	0	72	45	1	70	70.5	255	—	—	RNA 6913	—	—	—
	IR 657335	—	-0.015	73	35	0.6	69	71	235	—	—	—	—	NK 73/35	—
	IR 657530	—		75	30	1	70	73	260	—	—	—	—	—	RNAF 759530
	IR 657560	—		75	60	1	70	73	520	—	—	—	—	—	RNAFW 759560

INNER RINGS

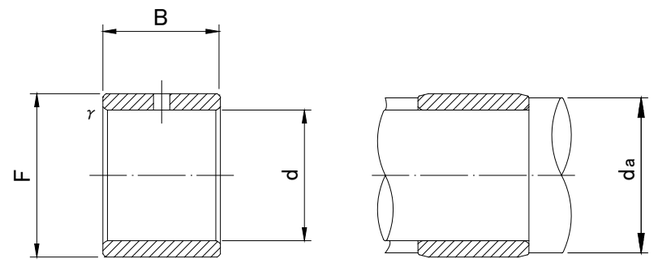






IR





IRZ

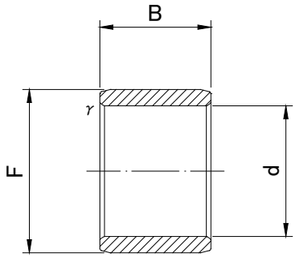
IR,IRZ TYPE

Shaft Diameter (mm)	Designation		Dimensions (mm)				Standard mounting dimensions (mm)		Mass	Usable bearing designation					
	IR	IRZ	d	F	B	*r*s min	da MIN	da MAX	g (approx)	RNA 49	RNA 59	RNA 69	RNA 48	NK	RNAF
70	IR 708025	—	70 0 -0.015	80	25	1	75	78	225	—	—	—	—	NK 80/25	—
	IR 708030	—		80	30	1	75	78	275	RNA 4914	—	—	—	—	RNAF 8010030
	IR 708035	—		80	35	1	75	78	310	—	—	—	—	NK 80/35	—
	IR 708040	—		80	40	1	75	78	358	—	RNA 5914	—	—	—	—
	IR 708054	—		80	54	1	75	78	490	—	—	RNA 6914	—	—	—
	IR 708060	—		80	60	1	75	78	560	—	—	—	—	—	RNAFW 8010060
75	IR 758525	—	75 0 -0.015	85	25	1	80	83	240	—	—	—	—	NK 85/25	—
	IR 758530	—		85	30	1	80	83	290	RNA 4915	—	—	—	—	RNAF 8510530
	IR 758535	—		85	35	1	80	83	335	—	—	—	—	NK 85/35	—
	IR 758540	—		85	40	1	80	83	385	—	RNA 5915	—	—	—	—
	IR 758554	—		85	54	1	80	83	520	—	—	RNA 6915	—	—	—
80	IR 809025	—	80 0 -0.015	90	25	1	85	88	255	—	—	—	—	NK 90/25	—
	IR 809030	—		90	30	1	85	88	310	RNA 4916	—	—	—	—	RNAF 9011030
	IR 809035	—		90	35	1	85	88	355	—	—	—	—	NK 90/35	—
	IR 809040	—		90	40	1	85	88	408	—	RNA 5916	—	—	—	—
	IR 809054	—		90	54	1	85	88	550	—	—	RNA 6916	—	—	—
85	IR 859526	—	85 0 -0.020	95	26	1	90	93	280	—	—	—	—	NK 95/26	—
	IR 859530	—		95	30	1	90	93	330	—	—	—	—	—	RNAF 9511530
	IR 859536	—		95	36	1	90	93	390	—	—	—	—	NK 95/36	—
	IR 8510035	—		100	35	1.1	91.5	98	575	RNA 4917	—	—	—	—	—
	IR 8510046	—		100	46	1.1	91.5	98	760	—	RNA 5917	—	—	—	—
	IR 8510063	—		100	63	1.1	91.5	98	1 040	—	—	RNA 6917	—	—	—
90	IR 9010026	—	90 0 -0.020	100	26	1	95	98	295	—	—	—	—	NK 100/26	—
	IR 9010030	—		100	30	1	95	98	355	—	—	—	—	—	RNAF 10012030
	IR 9010036	—		100	36	1	95	98	415	—	—	—	—	NK 100/36	—
	IR 9010535	—		105	35	1.1	96.5	103	610	RNA 4918	—	—	—	—	—
	IR 9010546	—		105	46	1.1	96.5	103	800	—	RNA 5918	—	—	—	—
	IR 9010563	—		105	63	1.1	96.5	103	1 100	—	—	RNA 6918	—	—	—
95	IR 9510526	—	95 0 -0.020	105	26	1	100	103	315	—	—	—	—	NK 105/26	—
	IR 9510536	—		105	36	1	100	103	430	—	—	—	—	NK 105/36	—
	IR 9511035	—		110	35	1.1	101.5	108	650	RNA 4919	—	—	—	—	—
	IR 9511046	—		110	46	1.1	101.5	108	850	—	RNA 5919	—	—	—	—
	IR 9511063	—		110	63	1.1	101.5	108	1 160	—	—	RNA 6919	—	—	—
100	IR 10011030	—	100 0 -0.020	110	30	1.1	106.5	108	380	—	—	—	—	NK 110/30	—
	IR 10011040	—		110	40	1.1	106.5	108	500	—	—	—	—	NK 110/40	—
	IR 10011540	—		115	40	1.1	106.5	113	700	RNA 4920	—	—	—	—	—

INNER RINGS

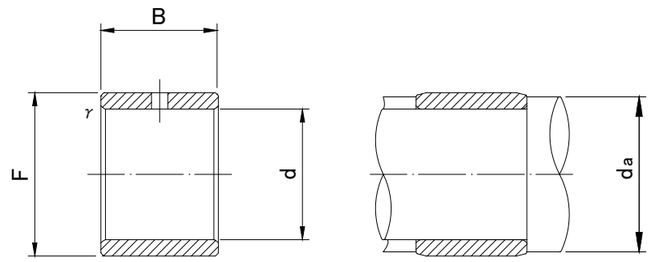






IR





IRZ

IR,IRZ TYPE

Shaft Diameter (mm)	Designation		Dimensions (mm)				Standard mounting dimensions (mm)		Mass	Usable bearing designation					
	IR	IRZ	d	F	B	*r*s min	da MIN	da MAX	g (approx)	RNA 49	RNA 59	RNA 69	RNA 48	NK	RNAF
110	**IR 11012030**	—	110 0 -0.020	120	30	1.0	115	118	410	—	—	—	RNA 4822	—	—
	IR 11012540	—	110 -0.020	125	40	1.1	116.5	123	840	RNA 4922	—	—	—	—	—
120	**IR 12013030**	—	120 0 -0.020	130	30	1.0	125	128	450	—	—	—	RNA 4824	—	—
	IR 12013545	—	120 -0.020	135	45	1.1	126.5	133	1 030	RNA 4924	—	—	—	—	—
130	**IR 13014535**	—	130 0 -0.025	145	35	1.1	136.5	143	860	—	—	—	RNA 4826	—	—
	IR 13015050	—	130 -0.025	150	50	1.5	138	148	1 670	RNA 4926	—	—	—	—	—
140	**IR 14015535**	—	140 0 -0.025	155	35	1.1	146.5	153	930	—	—	—	RNA 4828	—	—
	IR 14016050	—	140 -0.025	160	50	1.5	148	158	1 790	RNA 4928	—	—	—	—	—
150	**IR 15016540**	—	150 0 -0.025	165	40	1.1	156.5	163	1 130	—	—	—	RNA 4830	—	—
160	**IR 16017540**	—	160 0 -0.025	175	40	1.1	166.5	173	1 200	—	—	—	RNA 4832	—	—

CAM FOLLOWERS



Cam follower with pre-installed grease nipple

Newly available versatile cam follower which has conquered disadvantages in traditional products!!

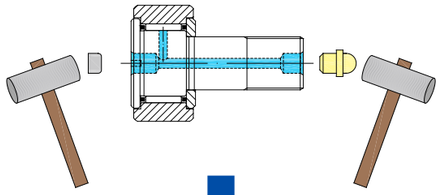
CF..AB Standard type | **CF..MAB** Stainless type | **CFH..AB** Eccentric type | **CFH..MAB** Eccentric stainless type

Ready use instantly after unpack the box thanks to pre-installed grease nipple.

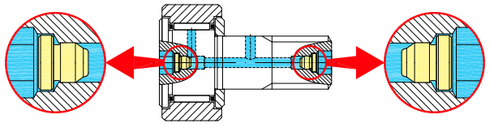


Lubrication available from both stud head side and thread side.No restriction for mounting location.

•Conventional type: Need to fit a nipple

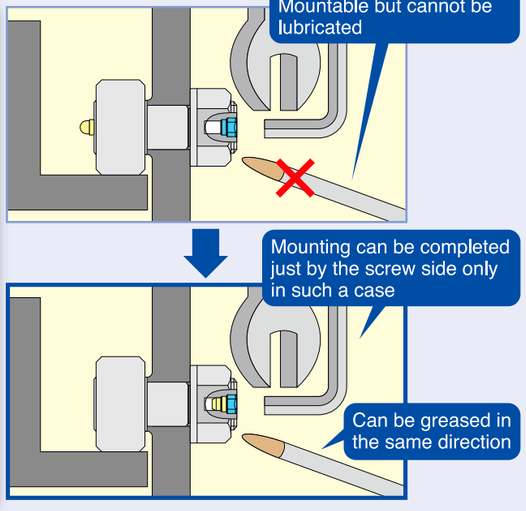


•Grease nipple fitted

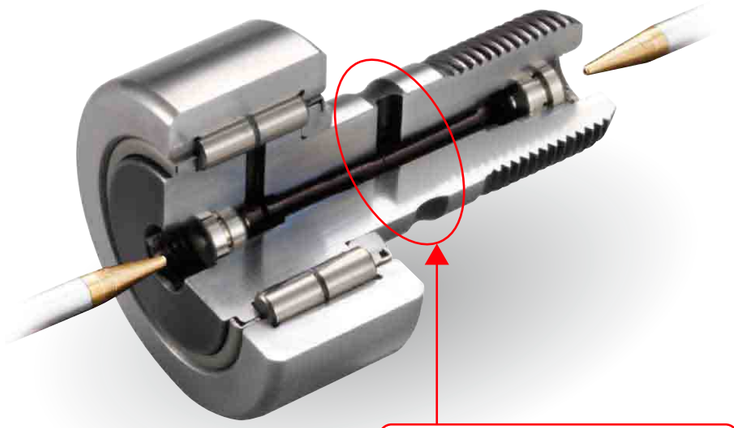


No restriction for mounting location thanks to hexagon socket shape in stud head and thread side.Easy mounting and removal is available thanks to having hexagon socket shape in both side.



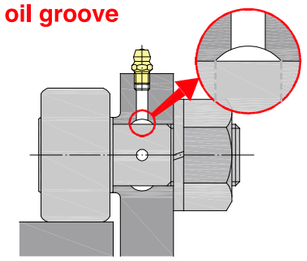


NUCF..AB Double-row cylindrical roller type | stud diameter **16 to 30mm**

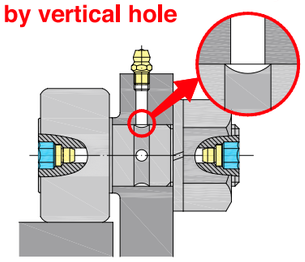


Lubrication area at the side of stud has an oil groove to eliminate bothersome groove machining process in the mounting side(NUCF type).

•Need to machine an internal oil groove

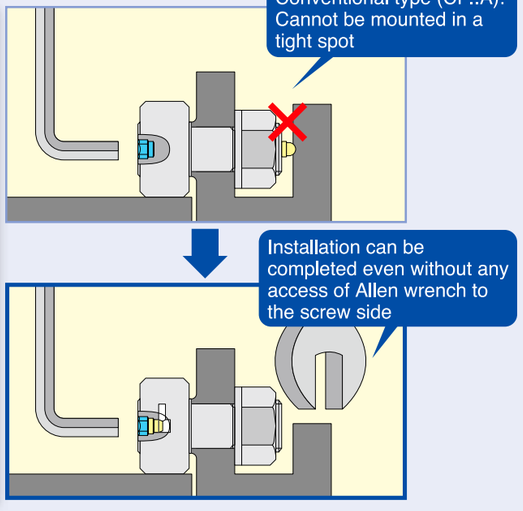


•Lubrication enabled only by vertical hole

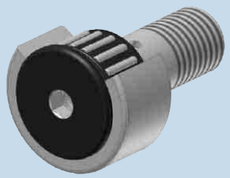
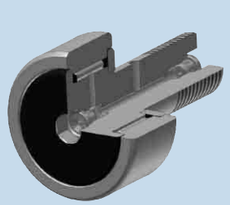
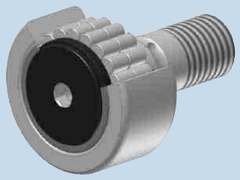
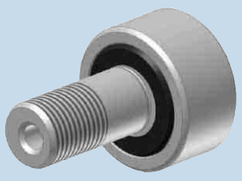


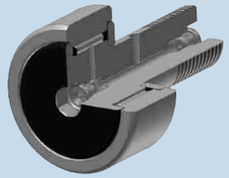
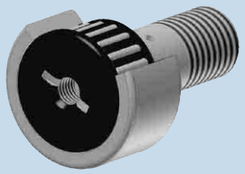
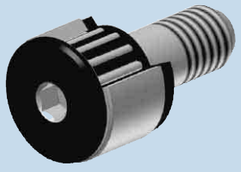
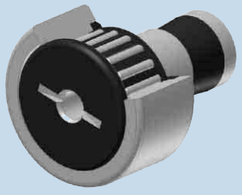
No restriction for mounting location thanks to hexagon socket shape in stud head and thread side.Easy mounting and removal is available thanks to having hexagon socket shape in both side.





Type and Part Code

Type	Applicable axis diameter	Feature	Part Code
 CF	ϕ 5 ~ ϕ 30	General purpose cam follower with screwdriver groove on the stud head. Available with stainless steel type (code M) for higher durability.	**V**: Full complement **None**: With cage **UU**: With seal **None**: With shield **CF** **16** **V** **M** **UU** **R** **N** Type **M**: Stainless steel **None**: High carbon steel **R**: Crowned outer ring **None**: Cylindrical outer ring Option (Dedicated grease nipple)
 CF-A Hexagon hole	ϕ 3 ~ ϕ 30	Hexagon sockets at the head of stud. Easy mounting with hex-wrench. Available with hexagon sockets at thread side model(CF-B). (Applicable for shaft diameter 12 or more) Available with stainless steel type (code M) for higher corrosive resistance.	**V**: Full complement **None**: With cage **UU**: With seal **None**: With shield **R**: Crowned outer ring **None**: Cylindrical outer ring **CF** **16** **V** **M** **UU** **R** **A** **N** Type **M**: Stainless steel **None**: High carbon steel **A**: with hexagon socket on the stud head **B**: with hexagon socket on the thread side (Applicable shaft diameter ϕ 12 ~ ϕ 30) Option (Dedicated grease nipple)
 CF-AB	ϕ 10 ~ ϕ 30	Both of stud head and thread ends have hexagon holes and integrated concave grease nipples. Available for mounting with either hexagon sockets and lubrication from both sides. Available with stainless steel type (code M) for higher corrosive resistance.	**V**: Full complement **None**: With cage **UU**: With seal **None**: With shield **R**: Crowned outer ring **None**: Cylindrical outer ring **CF** **16** **V** **M** **UU** **R** **AB** Type **M**: Stainless steel **None**: High carbon steel **AB**: with hexagon socket on both sides. (With integrated concave nipple)
 NUCF-AB Double cylindrical roller	ϕ 16 ~ ϕ 30	Cam follower which allows high radial loading and moderate axial loading with integrated cylindrical roller in double rows. Integrated concave grease nipples and hexagon sockets in both sides allow mounting and lubrication from either side.	**NUCF** **16** **R** **AB** Type **AB**: with hexagon socket on both sides. (With integrated concave nipple) **R**: Crowned outer ring **None**: Cylindrical outer rings
 CFH-A Hexagon socket, Eccentric	ϕ 5 ~ ϕ 30	Available for the same mounting hole as general type. Compact and high accuracy eccentric cam followers with integral structure enables easy fine positioning adjustment simply by rotating stud due to eccentric stud head shifting 0.2 to 1mm from stud mounting axis. Drastic reduction of machining and assembling process is enabled by eliminating positioning against cam groove or high accuracy machining of mounting holes. Available with stainless steel type (code M) for higher corrosive resistance.	**V**: Full complement **None**: With cage **UU**: With seal **None**: With shield **R**: Crowned outer ring **None**: Cylindrical outer ring **CFH** **16** **V** **M** **UU** **R** **A** Type **M**: Stainless steel **None**: High carbon steel **A**: with hexagon socket on the stud head **B**: with hexagon socket on the thread side (Applicable shaft diameter ϕ 12 ~ ϕ 30) **None**: Standard typed

Type	Applicable axis diameter	Feature	Part Code
 CFH-AB	ϕ 12 ~ ϕ 30	Eccentric cam follower with integrated concave grease nipples on both sides. Available for mounting and lubrication with hexagon holes on both sides. Available with stainless steel type (code M) for higher corrosive resistance.	**V**: Full complement **None**: With cage **UU**: With seal **None**: With shield **R**: Crowned outer ring **None**: Cylindrical outer ring **CFH 16 V M UU R AB** Type **M**: Stainless steel **None**: High carbon steel **AB**: with hexagon socket on both sides (Applicable shaft diameter ϕ 12~ ϕ 30) (With integrated concave nipple)
 CFT Lubrication tap hole	ϕ 6 ~ ϕ 30	Cam follower with tap for piping at stud head and thread of general type. Optimal for location that requires concentrated lubrication piping. Available with stainless steel type (code M) for higher corrosive resistance.	**V**: Full complement **None**: With cage **UU**: With seal **None**: With shield **R**: Crowned outer ring **None**: Cylindrical outer ring **CFT 16 V M UU R A** **CFT**: Standard typed **CFHT**: Eccentric type **M**: Stainless steel **None**: High carbon steel **A**: with hexagon socket on the stud head (Applicable shaft diameter ϕ 12 ~ ϕ 30) **B**: with hexagon socket on the thread side (Applicable shaft diameter ϕ 12 ~ ϕ 30) **None**: Standard typed
 CFS Compact outer ring	ϕ 2.5 ~ ϕ 6	Built-in very thin needle roller. Enables compact design due to very small outer ring diameter in comparison to stud diameter. Available with stainless steel type (code M) for higher corrosive resistance.	**CFS 6 V M A** Type **V**: Full complement **None**: With cage **M**: Stainless steel **None**: High carbon steel **A**: with hexagon socket on the stud head
 CF-SFU Easy mounting	ϕ 6 ~ ϕ 20	A type with step on stud to allow easy mounting. Optimal for devices with limited space for stud tightening. Available with seal only.	**V**: Full complement **None**: With cage **CF-SFU 16 V R** Type **R**: Crowned outer ring **None**: Cylindrical outer ring
 CR Inch type	ϕ 4.826 ~ ϕ 22.225	Cam followers in inch dimension.	**R**: Crowned outer ring **None**: Cylindrical outer ring **CR 16 UU R** Type **UU**: With seal **None**: With shield

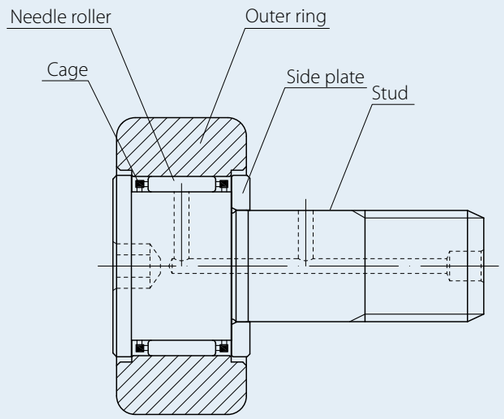
Structure and Features

Cam followers are compact and high rigid bearings with stud. It has a structure that an outer ring with built-in "roller" rolls over a track (raceway) and is utilized as guide roller for cam mechanism or linier motion. The cam follower is designed with a thick-walled outer ring in order to provide high rigidity to endure heavy load or impact load at contact portion with the track.

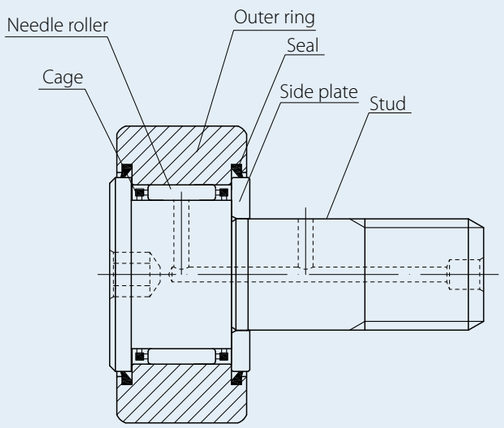
Outer rings have two shapes – crowned outer ring and cylindrical outer ring. The cylindrical outer ring is effective for heavy duty applications since it contacts with large contact area of the associated objects so that can mitigate the contact surface pressure. The crowned outer ring is suited for absorbing eccentric load since it absorbs angular misalignment due to a mount error. They have two internal structures – one with cage and one with full complement. Caged type cam followers are suited for high speed application, since the guiding feature of cage enables the rotation of "rollers" to be stable. Compared with caged type cam followers, full complement type is effective for applications with low-speed operation of heavy load due to its larger load rating. Full complement type also includes double-raw cylindrical roller type which allows moderate axial loading.

Cam followers shall be fixed by tightening hexagon nut at stud thread by holding the stud with screw driver or hex-wrench.

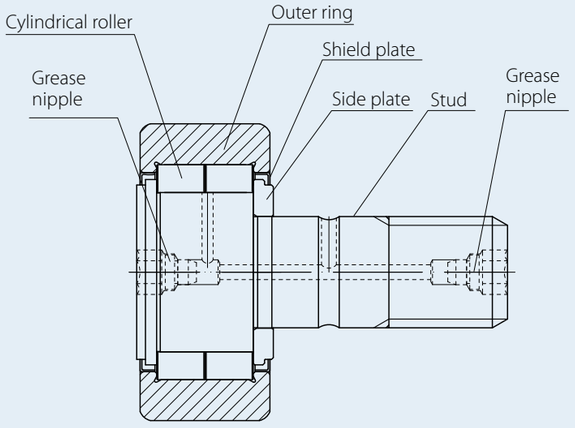
Cam followers with eccentric axis on the stud thread have an advantage to adjust the variation of mounting position within its eccentric range, without requesting high machining accuracy of mounting hole position.



CF··Structure of type A



CF··UU Structure of type A



NUCF··Structure of type AB

Accuracy standard

Accuracy of cam follower is indicated in Table-1 and Table-2 below.

Table-1 Accuracy

Unit: μm

Name \ Category	Cam follower series in metric (CF--, NUCF--)		Compact type (CFS--)	Cam follower series in inch (CR--)	
	Crowned outer ring	Cylindrical outer ring	Cylindrical outer ring	Crowned outer ring	Cylindrical outer ring
Dimension tolerance of outer ring outer diameter (D)	0 -50	Refer to Table-2	Refer to Table-2	0 -50	0 -25
Dimension tolerance of stud diameter (d)	h7		h6	+25 0	
Dimension tolerance of outer ring width (C)	0 -120		0 -120	0 -130	

Table-2 Accuracy of outer ring (Metric series and compact type roller outer ring)

Unit: μm

Nominal outer ring outer diameter D (mm)		Dimension tolerance of mean outside diameter in a single plane Δ_{Dmp}		Radial runout of outer ring K_{ea}
Over	Incl.	high	low	max.
6 (5 or more)	18	0	-8	15
18	30	0	-9	15
30	50	0	-11	20
50	80	0	-13	25
80	120	0	-15	35

Radial internal clearance

Table below indicates radial internal clearance of cam follower.

Table-3 Radial internal clearance

Unit: μm

Part code				Radial internal clearance	
Cam follower series in metric (CF)	Compact type (CFS)	Double-row cylindrical roller cam follower (NUCF)	Inch type series (CR)	min.	max.
CF3 ～ 5	CFS2.5 ～ 5		CR8 ～ 8-1	3	17
CF6 ～ 8	CFS6		CR10 ～ 10-2	5	20
CF10 ～ 12-1			CR12 ～ 22	5	25
CF16 ～ 20-1			CR24 ～ 26	10	30
CF24 ～ 30-2				10	40
		NUCF16 ～ 24		0	25
		NUCF24-1 ～ 30-2		5	30

Fits

Cam followers require machining of mounting hole to eliminate play at fitting portion especially for the portion which is subjected to impact shock due to its application in cantilevered mounting. Table-4 indicates recommended fits between cam follower and its mounting hole.

Table-4 Dimension tolerance of stud mounting hole

Part code	Dimension tolerance of stud mounting hole
Cam follower series in metric (CF)	H7
Compact type (CFS)	H6
Double-row cylindrical roller cam follower (NUCF)	H7
Cam follower series in inch (CR)	F7

Maximum permissive load

Permissive load of cam follower is subjected to change in accordance with bending strength and shear strength of its stud in addition to standard rating load of needle bearing due to its design with stud. This load is indicated as maximum permissive load.

Track load capacity

Track load capacity means a permissive load under which the outer ring of cam follower and the mating surface are allowable to be used over a long period without causing any deformation nor compression mark. Track load capacity depicted in dimension table indicates a value for which hardness of steel mating material is assumed to be HRC40. In the case that the hardness of mating material is not HRC40, track load capacity in the dimension table shall be multiplied by value of track load factor obtained by Figure-1.

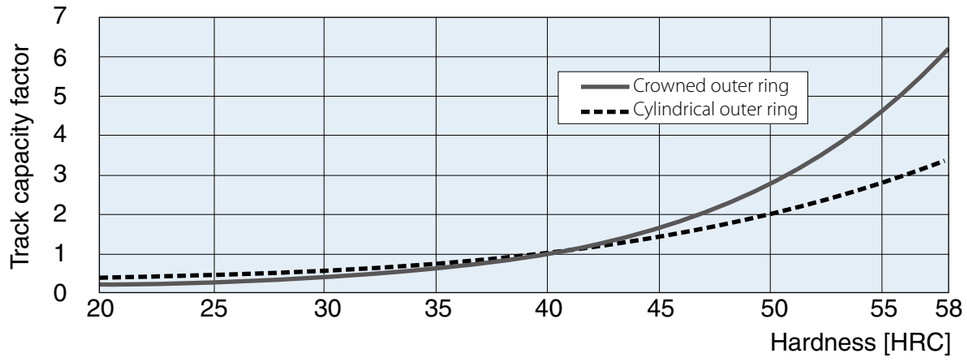


Figure-1 Track capacity factor

Lubrication

All the JNS cam followers are lubricated with pre-packed high quality lithium soap-based grease grade 2 (RoHS compliant) so that they are ready to use. For applications requiring high prevention measure against invasion of foreign matter or leakage of lubricant, products with seal(part conde--UU) which integrate special synthetic lubber with high abrasion resistance are also available.

Grease shall be packed up to volume approximately one-third to one-half of internal space of bearing. Lubrication interval varies depending on operation condition. Referential interval may be every six to twenty four months for cam followers with cages and every one to six months for full complement type with grease in the same type.

Some excessive grease may leak at the beginning of usage or immediately after re-greasing even for the products with seal. Aging operation period is recommended prior to application in which no contamination by grease is allowed in and around device. Wipe and clean any leaked extra grease after this operation.

Dedicated grease nipple shall be knocked in onto cam follower by pressing flange portion of the nipple using fixture shown in Figure-2.

Accessories

Accessories for cam follower of standard specification are shown in Table-5. Dedicated grease nipple is available upon customer request by ordering products with suffix "N" added to part code.

Also, type CF-AB and type NUCF-AB have integrated dedicated concave grease nipples installed at both sides.

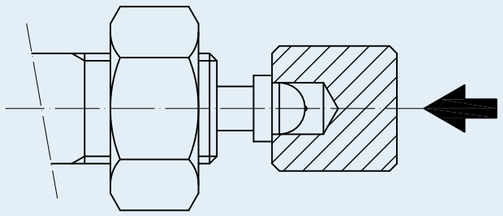


Figure-2

Example) CF 12 UUR -N
└ Dedicated grease nipple

Table-5 Accessories

Part code		Stop plug*1)	Resin plug*2)	Hexagon nuts style 2	Grease
CF CFH CR	Without seal	Attached	Attached	Attached	Packed
	With seal	Attached	Attached	Attached	Packed
CFT	Without seal	—	—	Attached	Packed
	With seal	—	—	Attached	Packed
CFS		—	—	Attached	Packed
CF-SFU		Attached	Attached	—	Packed
NUCF		—	—	Attached	Packed

*1) Stop plug is used for plugging unused lubrication hole(s).
*2) Resin plug is used for preventing leakage of grease.

DIMENSIONS OF GREASE NIPPLES

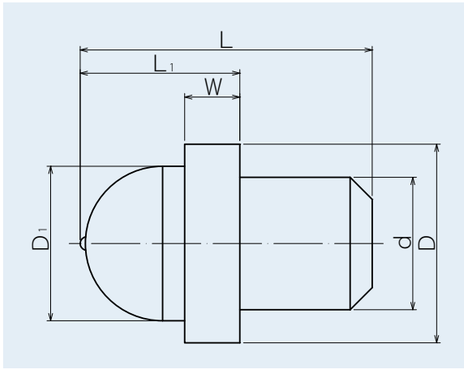


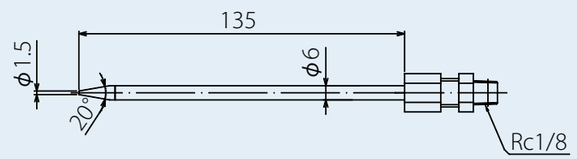
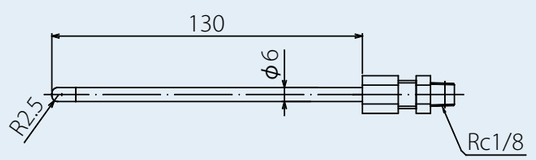
Table-6

Applicable part code	Nipple dimension					
CF, CFH	d	D	D_1	L	L_1	W
CF5	3.2	7.5	6	9	5.5	1.5
CF6 ～ CF10-1	4	7.5	6	10	5.5	1.5
CF12 ～ CF18	6	8	6	11	6	2
CF20 ～ CF30-2	8	10	6	16	7	3

Dimensions of supply nozzle for CF..AB type



Table-7 Dimension of recommended lubrication nozzle (for CF..AB, NUCF..AB)

Type	Dimension and shape	Applied model number	
NPAB-1		CF10ABK(M10 × 1) CF10-1ABK(M10 × 1) CF12AB CF12-1AB CF16AB CF18AB	NUCF16AB NUCF18AB
NPAB-2		CF20AB CF20-1AB CF24AB CF24-1AB CF30AB CF30-1AB CF30-2AB	NUCF20AB NUCF20-1AB NUCF24AB NUCF24-1AB NUCF30AB NUCF30-2AB

Mounting

Mounting part

Hole for stud shall be perpendicular to mounting surface so as to ensure even contact between outer ring and rolling surface of mating material. Corner chamfer of hole shall be as small as around C0.5. Counter bore diameter shall be dimension f or more as described in the dimension table.

Type CF-R with crowned outer ring is recommended in the case of application with insufficient contact between outer ring and rolling surface of mating material.

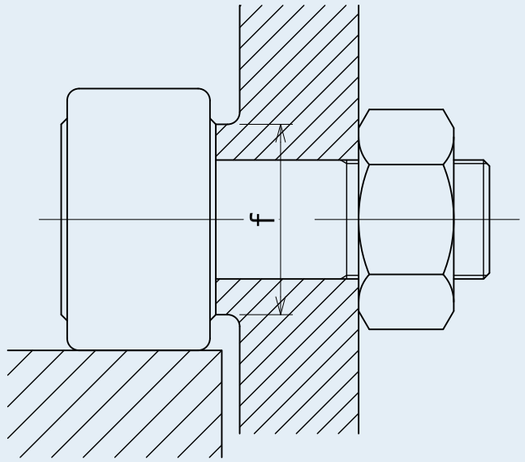


Figure-3

Mounting method

- Cam follower shall be mounted so as not to incline against direction of the motion.
- Mounting method to tighten cam follower directly to bracket by adding female thread to it without using nut as depicted in Figure-4 (A) is NOT recommended. This method may cause damage on stud due to concentration of bending stress in male thread portion in the event of loosening since it is difficult to tighten the stud sufficiently.

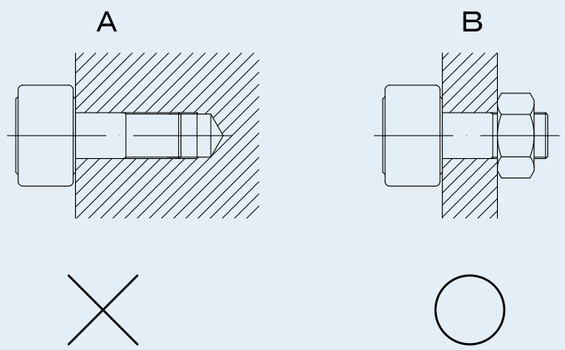


Figure-4

- For application with heavy load, lubrication hole of stud shall be located outside of loading range (on the side to receive load). Location of the lubrication hole is indicated by “JNS” marking on the side of stud flange. (Refer to Figure-5)
  Hole at the center of stud is used as stopper for tightening or lubricant supply hole for grease lubrication.

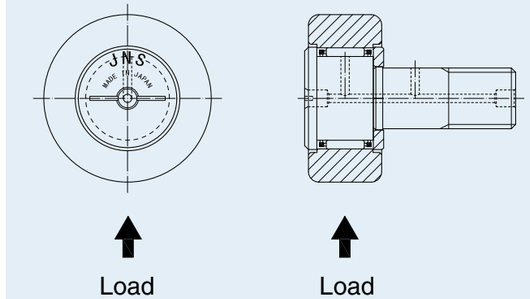


Figure-5

Precaution for using spring washer

It is important to ensure that spring washer used for mounting cam follower has no burr or sharp edge. Debris scraped from nut or mounting bracket by burr or shape edge of washer during tightening can invade stud thread and can cause insufficient tightness or damage to thread.

Tightening torque of stud

Stud of cam follower is subjected to bending stress or tensile stress by bearing load. Tightening torque shall be set not to exceed the value in the dimension table.

Using double spring washers, double thin nut of JIS B 1181 or special nut with locking feature is recommended in the case of possible loosening of mounting screw due to vibration or impact shock.

Mounting of eccentric cam follower

Adjustment of eccentric shall be performed in accordance with the following procedure.

(1) Insert stud to mounting hole and tighten nut as the stud can turn loosely. Marking of stud shall face location indicated in Figure-6 in relation to direction of load.

(2) Gap between cam follower and mating contact surface may be adjusted by turning stud using hexagon hole on the stud head.

(3) After completing adjustment, tighten nut with holding rotation of stud. Make it sure not to exceed maximum tightening torque of the nut.

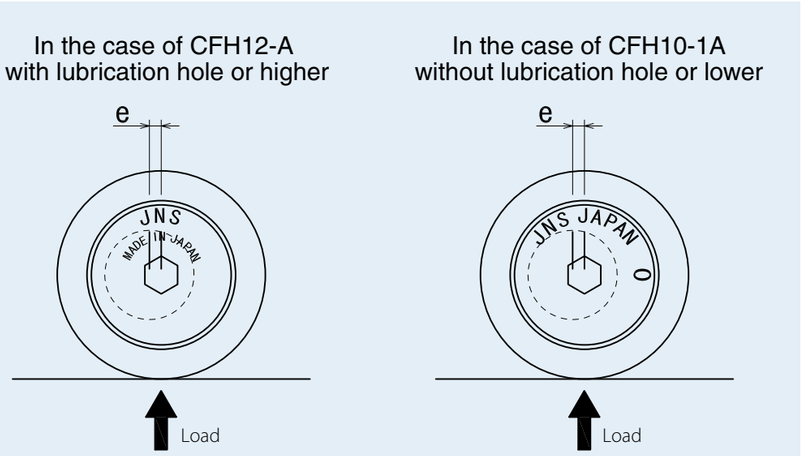


Figure-6

Mounting of CF-SFU type

Refer to Figure-7 for mounting method of easy mounting of CF-SFU type.

Type CF-SFU is NOT recommended for application in the part subjected to vibration or shock impact as much as its mounting is simple. Standard cam followers with nut are recommended for application with possible vibration or shock impact.

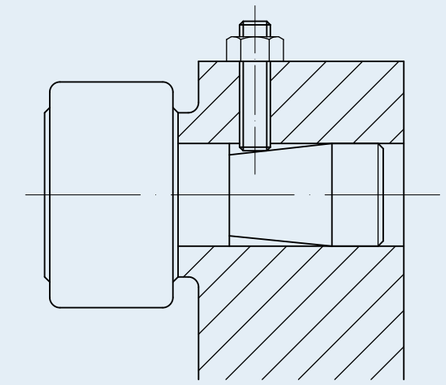
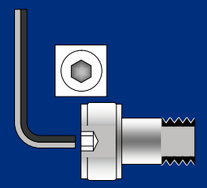


Figure-7

CAM FOLLOWERS
MINIATURE TYPE
HEXAGON SOCKET ON STUD HEAD

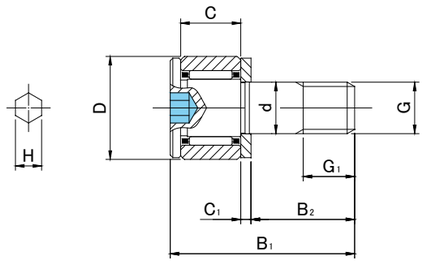


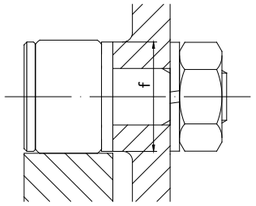


CFS..A



CFS..VA





CFS..A TYPE

Prepacked Grease

Stud diameter (mm)	h6 tolerance	Designation Cylindrical outer ring: With cage	Full roller	D	C	d	G	G1	B1	B2	C1	H	f min	Basic dynamic load rating Cr N	Basic static load rating Cor N	Largest permissible load N	Max tightening torque N·m	Mass g (approx)
2.5	0 -0.006	**CFS 2.5A**	—	**5**	3	2.5	M2.5×0.45	2.5	9.5	5	0.7	0.9	4.8	410	335	260	0.2	1
		—	**CFS 2.5VA**											1 000	1 080	260		
3	0 -0.006	**CFS 3A**	—	**6**	4	3	M3×0.5	3	11.5	6	0.7	1.5	5.8	630	610	360	0.3	2
		—	**CFS 3VA**											1 370	1 770	360		
4	0 -0.008	**CFS 4A**	—	**8**	5	4	M4×0.7	4	15	8	1	2	7.7	1 080	1 080	780	0.6	4
		—	**CFS 4VA**											2 350	3 040	780		
5	0 -0.008	**CFS 5A**	—	**10**	6	5	M5×0.8	5	18	10	1	2.5	9.6	1 570	1 860	1 420	1.3	7
		—	**CFS 5VA**											3 140	4 710	1 420		
6	0 -0.008	**CFS 6A**	—	**12**	7	6	M6×1	6	21.5	12	1.2	3	11.6	2 060	2 160	2 110	2.3	13
		—	**CFS 6VA**											4 610	6 270	2 110		

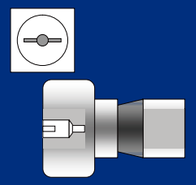
OUTER RINGS TOLERANCE (μm)

TYPE	Cylindrical outer ring
CFS2.5,CFS3,CFS4,CFS5,CFS6	0/-8

ACCESSORIES

TYPE	 NUT
All types	Attached

CAM FOLLOWERS
EASY MOUNTING TYPE
SCREWDRIVER SLOT HEAD

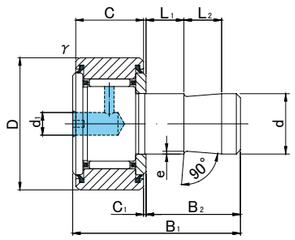


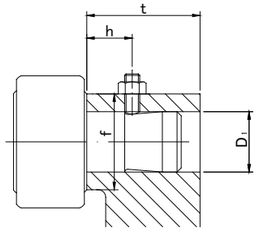


CF-SFU



CF-SFU..V





CF-SFU TYPE

Prepacked Grease

Stud diameter (mm)		Designation R500(CF6~CF18) R1000(CF20≦)		Dimensions (mm)										Mounting dimensions (mm)					Basic dynamic load rating	Basic static load rating	Largest permissible load	Track load capacity		Mass
	h7 tolerance	Cylindrical outer ring: With seals	Crowned outer ring: With seals	D	C	d	B1	B2	C1	L1	L2	d1	e	rs min	D1 Tolerance	t Min	f Min	h (Ref)	Cr N	Cor N	N	Cylindrical outer ring N	Crowned outer ring N	g (approx)
6	0 -0.012	CF-SFU-6	CF-SFU-6R	16	11	6	32	20	0.6	5	10	4	0.3	0.3	6 0 +0.012	20	11	10	3 630	3 630	2 110	3 430	1 080	19
		CF-SFU-6V	CF-SFU-6VR																6 960	8 530	2 110	3 430	1 080	
8	0 -0.015	CF-SFU-8	CF-SFU-8R	19	11	8	32	20	0.6	5	10	4	0.5	0.3	8 0 +0.015	20	13	10	4 310	4 710	4 710	4 020	1 370	28.5
		CF-SFU-8V	CF-SFU-8VR																8 130	11 170	4 710	4 020	1 370	
10	0 -0.015	CF-SFU-10	CF-SFU-10R	22	12	10	33	20	0.6	5	10	4	0.5	0.3	10 0 +0.015	20	15	10	5 390	6 860	6 860	4 700	1 670	43
		CF-SFU-10V	CF-SFU-10VR																9 510	14 500	7 450	4 700	1 670	
10	0 -0.015	CF-SFU-10-1	CF-SFU-10-1R	26	12	10	33	20	0.6	5	10	4	0.5	0.3	10 0 +0.015	20	15	10	5 390	6 860	6 860	5 490	2 060	58.5
		CF-SFU-10-1V	CF-SFU-10-1VR																9 510	14 500	7 450	5 490	2 060	
12	0 -0.018	CF-SFU-12	CF-SFU-12R	30	14	12	35	20	0.6	5	10	6	1	0.6	12 0 +0.018	20	20	10	7 940	9 800	9 800	7 060	2 450	93
		CF-SFU-12V	CF-SFU-12VR																13 430	19 700	11 270	7 060	2 450	
12	0 -0.018	CF-SFU-12-1	CF-SFU-12-1R	32	14	12	35	20	0.6	5	10	6	1	0.6	12 0 +0.018	20	20	10	7 940	9 800	9 800	7 450	2 740	103
		CF-SFU-12-1V	CF-SFU-12-1VR																13 430	19 700	11 270	7 450	2 740	
16	0 -0.018	CF-SFU-16	CF-SFU-16R	35	18	16	44.5	25	0.8	10	10	6	1	0.6	16 0 +0.018	25	24	15	12 050	18 330	18 330	11 200	3 140	164
		CF-SFU-16V	CF-SFU-16VR																20 680	37 630	19 800	11 200	3 140	
18	0 -0.018	CF-SFU-18	CF-SFU-18R	40	20	18	46.5	25	0.8	10	10	6	1	1	18 0 +0.018	25	26	15	14 700	25 200	25 200	14 400	3 720	235
		CF-SFU-18V	CF-SFU-18VR																25 280	51 350	26 560	14 400	3 720	
20	0 -0.021	CF-SFU-20	CF-SFU-20R	52	24	20	50.5	25	0.8	10	10	8	1	1	20 0 +0.021	25	36	15	20 680	34 600	32 140	23 200	8 230	436
		CF-SFU-20V	CF-SFU-20VR																33 120	64 480	32 140	23 200	8 230	
20	0 -0.021	CF-SFU-20-1	CF-SFU-20-1R	47	24	20	50.5	25	0.8	10	10	8	1	1	20 0 +0.021	25	36	15	20 680	34 600	32 140	21 000	7 150	361
		CF-SFU-20-1V	CF-SFU-20-1VR																32 120	64 480	32 140	21 000	7 150	

OUTER RINGS TOLERANCE (μm)

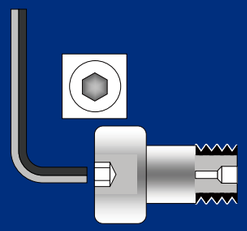
TYPE	Cylindrical outer ring	Crowned outer ring
CF6	0/-8	0/-50
CF8,CF10,CF10-1,CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20	0/-13	0/-50

ACCESSORIES

TYPE	 STOP PLUG	 RESIN PLUG
CF6/CF8/CF10/CF10-1	—	ϕ 4 Attached
CF12/CF12-1/CF16/CF18	ϕ 6 Attached	ϕ 6 Attached
CF20/CF20-1	ϕ 8 Attached	ϕ 8 Attached

CF-SFU

CAM FOLLOWERS
HEXAGON SOCKET ON STUD HEAD

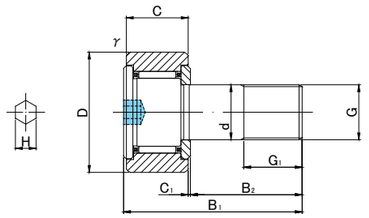




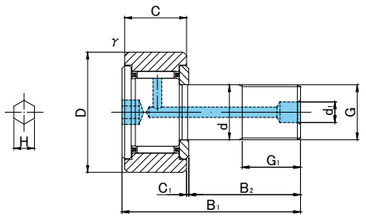
CF..A



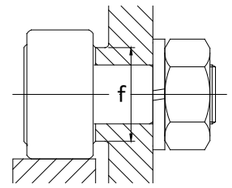
CF..VA



CF3(V)A～CF10−1(V)A



CF10(V)AK～CF10−1(V)AK



CF..A TYPE

Prepacked Grease

Stud diameter (mm)		Designation: Cylindrical outer ring		Designation: Crowned outer ring R250(≦CF5) R500(CF6～CF10)		Dimensions (mm)														Basic dynamic load rating	Basic static load rating	Largest permissible load	Limiting speed *	Track load capacity		Max tightening torque	Mass
	h7 tolerance	Without seals	With seals	Without seals	With seals	D	C	d	G	G1	B1	B2	B3	C1	d1	d2	H	r_s min	f min	Cr N	Cor N	N	rpm	Cylindrical outer ring N	Crowned outer ring N	N·m	g (approx)
3	0 -0.010	**CF 3A**	**CF 3UUA**	**CF 3RA**	**CF 3UURA**	**10**	7	3	M3×0.5	5	17	9	—	0.5	—	—	2	0.2	6.8	1 470	1 180	360	47 000	1 370	540	0.4	4.5
		CF 3VA	**CF 3VUUA**	**CF 3VRA**	**CF 3VUURA**															2 800	2 500	360	18 800				
4	0 -0.012	**CF 4A**	**CF 4UUA**	**CF 4RA**	**CF 4UURA**	**12**	8	4	M4×0.7	6	20	11	—	0.5	—	—	2.5	0.3	8.6	2 060	2 050	780	37 000	1 760	690	1	7.5
		CF 4VA	**CF 4VUUA**	**CF 4VRA**	**CF 4VUURA**															4 000	4 300	780	14 800				
5	0 -0.012	**CF 5A**	**CF 5UUA**	**CF 5RA**	**CF 5UURA**	**13**	9	5	M5×0.8	7.5	23	13	—	0.5	—	—	3	0.3	9.7	3 140	2 770	1 420	29 000	2 250	780	2	10.5
		CF 5VA	**CF 5VUUA**	**CF 5VRA**	**CF 5VUURA**															5 100	5 500	1 420	11 600				
6	0 -0.012	**CF 6A**	**CF 6UUA**	**CF 6RA**	**CF 6UURA**	**16**	11	6	M6×1	8	28	16	—	0.6	—	—	3	0.3	11	3 630	3 630	2 110	25 000	3 430	1 080	3	18.5
		CF 6VA	**CF 6VUUA**	**CF 6VRA**	**CF 6VUURA**															6 960	8 530	2 110	12 000				
8	0 -0.015	**CF 8A**	**CF 8UUA**	**CF 8RA**	**CF 8UURA**	**19**	11	8	M8×1.25	10	32	20	—	0.6	—	—	4	0.3	13	4 310	4 710	4 710	20 000	4 020	1 370	8	28.5
		CF 8VA	**CF 8VUUA**	**CF 8VRA**	**CF 8VUURA**															8 130	11 170	4 710	9 000				
10	0 -0.015	**CF10A**	**CF10UUA**	**CF10RA**	**CF10UURA**	**22**	12	10	M10×1.25	12	36	23	—	0.6	—	—	5	0.3	15	5 390	6 860	6 860	17 000	4 700	1 670	15	45
		CF10VA	**CF10VUUA**	**CF10VRA**	**CF10VUURA**															9 510	14 500	7 450	7 500				
10	0 -0.015	**CF10-1A**	**CF10-1UUA**	**CF10-1RA**	**CF10-1UURA**	**26**	12	10	M10×1.25	12	36	23	—	0.6	—	—	5	0.3	15	5 390	6 860	6 860	17 000	5 490	2 060	15	60
		CF10-1VA	**CF10-1VUUA**	**CF10-1VRA**	**CF10-1VUURA**															9 510	14 500	7 450	7 500				
10	0 -0.015	**CF10AK**	**CF10UUAK**	**CF10RAK**	**CF10UURAK**	**22**	12	10	M10×1	12	36	23	—	0.6	4	—	5	0.3	15	5 390	6 860	6 860	17 000	4 700	1 670	15	45
		CF10VAK	**CF10VUUAK**	**CF10VRAK**	**CF10VUURAK**															9 510	14 500	7 450	7 500				
10	0 -0.015	**CF10-1AK**	**CF10-1UUAK**	**CF10-1RAK**	**CF10-1UURAK**	**26**	12	10	M10×1	12	36	23	—	0.6	4	—	5	0.3	15	5 390	6 860	6 860	17 000	5 490	2 060	15	60
		CF10-1VAK	**CF10-1VUUAK**	**CF10-1VRAK**	**CF10-1VUURAK**															9 510	14 500	7 450	7 500				

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

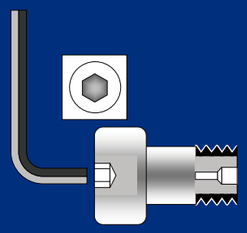
OUTER RINGS TOLERANCE (μm)

TYPE	Cylindrical outer ring	Crowned outer ring
CF3,CF4,CF5,CF6	0/-8	0/-50
CF8,CF10,CF10-1,CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	 STOP PLUG	 RESIN PLUG	 NUT
CF3/CF4/CF5/CF6/CF8/CF10/CF10-1	—	—	Attached
CF10K/CF10-1K	ϕ 4 Attached	ϕ 4 Attached	Attached
CF12/CF12-1/CF16/CF18	ϕ 6 Attached	ϕ 6 Attached	Attached
CF20/CF20-1/CF24/CF24-1/CF30/CF30-1/CF30-2	ϕ 8 Attached	ϕ 8 Attached	Attached

CAM FOLLOWERS
HEXAGON SOCKET ON STUD HEAD

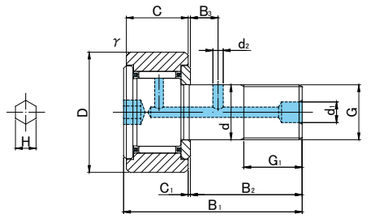




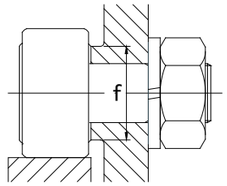
CF..A



CF..VA



CF12(V)A～CF30-2(V)A



CF..A TYPE

Prepacked Grease

Stud diameter (mm)	h7 tolerance	Designation: Cylindrical outer ring, Without seals	Designation: Cylindrical outer ring, With seals	Designation: Crowned outer ring R500(CF12～CF18) R1000(CF20≦), Without seals	Designation: Crowned outer ring R500(CF12～CF18) R1000(CF20≦), With seals	D	C	d	G	G1	B1	B2	B3	C1	d1	d2	H	r_s min	f min	Basic dynamic load rating Cr N	Basic static load rating Cor N	Largest permissible load N	Limiting speed * rpm	Track load capacity: Cylindrical outer ring N	Track load capacity: Crowned outer ring N	Max tightening torque N·m	Mass g (approx)
12	0 -0.018	**CF12A**	**CF12UUA**	**CF12RA**	**CF12UURA**	**30**	14	12	M12×1.5	13	40	25	6	0.6	6	3	6	0.6	20	7 940	9 800	9 800	14 000	7 060	2 450	22	95
		CF12VA	**CF12VUUA**	**CF12VRA**	**CF12VUURA**															13 430	19 700	11 270	6 000				
12	0 -0.018	**CF12-1A**	**CF12-1UUA**	**CF12-1RA**	**CF12-1UURA**	**32**	14	12	M12×1.5	13	40	25	6	0.6	6	3	6	0.6	20	7 940	9 800	9 800	14 000	7 450	2 740	22	105
		CF12-1VA	**CF12-1VUUA**	**CF12-1VRA**	**CF12-1VUURA**															13 430	19 700	11 270	6 000				
16	0 -0.018	**CF16A**	**CF16UUA**	**CF16RA**	**CF16UURA**	**35**	18	16	M16×1.5	17	52	32.5	8	0.8	6	3	6	0.6	24	12 050	18 330	18 330	10 000	11 200	3 140	58	170
		CF16VA	**CF16VUUA**	**CF16VRA**	**CF16VUURA**															20 680	37 630	19 800	4 500				
18	0 -0.018	**CF18A**	**CF18UUA**	**CF18RA**	**CF18UURA**	**40**	20	18	M18×1.5	19	58	36.5	8	0.8	6	3	6	1	26	14 700	25 200	25 200	8 500	14 400	3 720	87	250
		CF18VA	**CF18VUUA**	**CF18VRA**	**CF18VUURA**															25 280	51 350	26 560	3 500				
20	0 -0.021	**CF20A**	**CF20UUA**	**CF20RA**	**CF20UURA**	**52**	24	20	M20×1.5	21	66	40.5	9	0.8	8	4	8	1	36	20 680	34 600	32 140	7 000	23 200	8 230	120	460
		CF20VA	**CF20VUUA**	**CF20VRA**	**CF20VUURA**															33 120	64 480	32 140	3 500				
20	0 -0.021	**CF20-1A**	**CF20-1UUA**	**CF20-1RA**	**CF20-1URA**	**47**	24	20	M20×1.5	21	66	40.5	9	0.8	8	4	8	1	36	20 680	34 600	32 140	7 000	21 000	7 150	120	385
		CF20-1VA	**CF20-1VUUA**	**CF20-1VRA**	**CF20-1VURA**															33 120	64 480	32 140	3 500				
24	0 -0.021	**CF24A**	**CF24UUA**	**CF24RA**	**CF24UURA**	**62**	29	24	M24×1.5	25	80	49.5	11	0.8	8	4	8	1	40	30 480	52 630	49 500	6 500	34 200	10 500	220	815
		CF24VA	**CF24VUUA**	**CF24VRA**	**CF24VURA**															46 550	92 020	49 500	3 000				
24	0 -0.021	**CF24-1A**	**CF24-1UUA**	**CF24-1RA**	**CF24-1URA**	**72**	29	24	M24×1.5	25	80	49.5	11	0.8	8	4	8	1	40	30 480	52 630	49 500	6 500	39 800	12 900	220	1 140
		CF24-1VA	**CF24-1VUUA**	**CF24-1VRA**	**CF24-1VURA**															46 550	92 020	49 500	3 000				
30	0 -0.021	**CF30A**	**CF30UUA**	**CF30RA**	**CF30URA**	**80**	35	30	M30×1.5	32	100	63	15	1	8	4	8	1	46	45 370	85 060	73 700	5 000	52 600	14 900	450	1 870
		CF30VA	**CF30VUUA**	**CF30VRA**	**CF30VURA**															67 620	144 060	73 700	2 200				
30	0 -0.021	**CF30-1A**	**CF30-1UUA**	**CF30-1RA**	**CF30-1URA**	**85**	35	30	M30×1.5	32	100	63	15	1	8	4	8	1	46	45 370	85 060	73 700	5 000	56 000	56 000	450	2 030
		CF30-1VA	**CF30-1VUUA**	**CF30-1VRA**	**CF30-1VURA**															67 620	144 060	73 700	2 200				
30	0 -0.021	**CF30-2A**	**CF30-2UUA**	**CF30-2RA**	**CF30-2URA**	**90**	35	30	M30×1.5	32	100	63	15	1	8	4	8	1	46	45 370	85 060	73 700	5 000	59 300	59 300	450	2 220
		CF30-2VA	**CF30-2VUUA**	**CF30-2VRA**	**CF30-2VURA**															67 620	144 060	73 700	2 200				

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

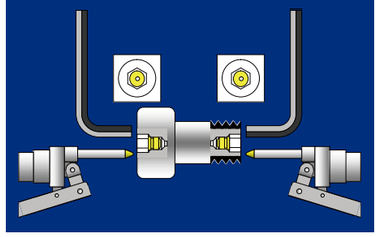
OUTER RINGS TOLERANCE (μm)

TYPE	Cylindrical outer ring	Crowned outer ring
CF3,CF4,CF5,CF6	0/-8	0/-50
CF8,CF10,CF10-1,CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	STOP PLUG	RESIN PLUG	NUT
CF3/CF4/CF5/CF6/CF8/CF10/CF10-1	—	—	Attached
CF10K/CF10-1K	ϕ 4 Attached	ϕ 4 Attached	Attached
CF12/CF12-1/CF16/CF18	ϕ 6 Attached	ϕ 6 Attached	Attached
CF20/CF20-1/CF24/CF24-1/CF30/CF30-1/CF30-2	ϕ 8 Attached	ϕ 8 Attached	Attached

CAM FOLLOWERS
HEXAGON SOCKET ON BOTH SIDES GREASE NIPPLE INSTALLED

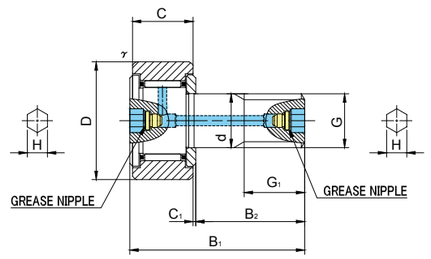




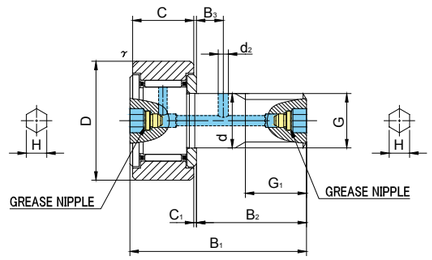
CF..AB



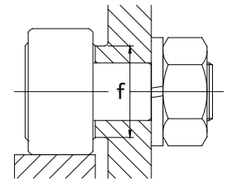
CF..VAB



CF10(V)ABK～CF10-1(V)ABK



CF12(V)AB～CF18(V)AB



CF..AB TYPE

Prepacked Grease

Stud diameter (mm)	h7 tolerance	Designation: Cylindrical outer ring, Without seals	Cylindrical outer ring, With seals	Crowned outer ring R500(CF10～CF18) R1000(CF20≦), Without seals	Crowned outer ring, With seals	D	C	d	G	G1	B1	B2	B3	C1	d2	H	rs min	f min	Basic dynamic load rating Cr N	Basic static load rating Cor N	Largest permissible load N	Limiting speed * rpm	Track load capacity: Cylindrical outer ring N	Track load capacity: Crowned outer ring N	Max tightening torque N·m	Mass g (approx)
10	0 -0.015	**CF10ABK**	**CF10UUABK**	**CF10RABK**	**CF10UURABK**	**22**	12	10	**M10×1**	12	36	23	—	0.6	—	5	0.3	15	5 390	6 860	6 860	17 000	4 700	1 670	15	45
		CF10VABK	**CF10VUUABK**	**CF10VRABK**	**CF10VUURABK**														9 510	14 500	7 450	7 500				
10	0 -0.015	**CF10-1ABK**	**CF10-1UUABK**	**CF10-1RABK**	**CF10-1UURABK**	**26**	12	10	**M10×1**	12	36	23	—	0.6	—	5	0.3	15	5 390	6 860	6 860	17 000	5 490	2 060	15	60
		CF10-1VABK	**CF10-1VUUABK**	**CF10-1VRABK**	**CF10-1VUURABK**														9 510	14 500	7 450	7 500				
12	0 -0.018	**CF12AB**	**CF12UUAB**	**CF12RAB**	**CF12UURAB**	**30**	14	12	M12×1.5	13	40	25	6	0.6	3	6	0.6	20	7 940	9 800	9 800	14 000	7 060	2 450	22	95
		CF12VAB	**CF12VUUAB**	**CF12VRAB**	**CF12VUURAB**														13 430	19 700	11 270	6 000				
12	0 -0.018	**CF12-1AB**	**CF12-1UUAB**	**CF12-1RAB**	**CF12-1UURAB**	**32**	14	12	M12×1.5	13	40	25	6	0.6	3	6	0.6	20	7 940	9 800	9 800	14 000	7 450	2 740	22	105
		CF12-1VAB	**CF12-1VUUAB**	**CF12-1VRAB**	**CF12-1VUURAB**														13 430	19 700	11 270	6 000				
16	0 -0.018	**CF16AB**	**CF16UUAB**	**CF16RAB**	**CF16UURAB**	**35**	18	16	M16×1.5	17	52	32.5	8	0.8	3	6	0.6	24	12 050	18 330	18 330	10 000	11 200	3 140	58	170
		CF16VAB	**CF16VUUAB**	**CF16VRAB**	**CF16VUURAB**														20 680	37 630	19 800	4 500				
18	0 -0.018	**CF18AB**	**CF18UUAB**	**CF18RAB**	**CF18UURAB**	**40**	20	18	M18×1.5	19	58	36.5	8	0.8	3	6	1	26	14 700	25 200	25 200	8 500	14 400	3 720	87	250
		CF18VAB	**CF18VUUAB**	**CF18VRAB**	**CF18VUURAB**														25 280	51 350	26 560	3 500				
20	0 -0.021	**CF20AB**	**CF20UUAB**	**CF20RAB**	**CF20UURAB**	**52**	24	20	M20×1.5	21	66	40.5	9	0.8	4	8	1	36	20 680	34 600	32 140	7 000	23 200	8 230	120	460
		CF20VAB	**CF20VUUAB**	**CF20VRAB**	**CF20VUURAB**														33 120	64 480	32 140	3 500				

CF..AB TYPE has oil holes (grease nipple) which are prepared in the stud head and thread side, also an oil hole is prepared in the outer surface of stud.

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

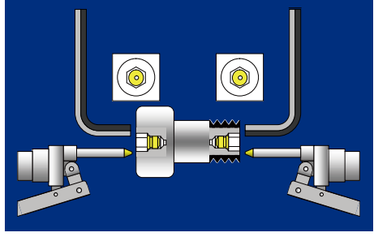
OUTER RINGS TOLERANCE (μm)

TYPE	Cylindrical outer ring	Crowned outer ring
CF10,CF10-1,CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	Grease Nipple	NUT
All types	Installed	Attached

CAM FOLLOWERS
HEXAGON SOCKET ON BOTH SIDES GREASE NIPPLE INSTALLED

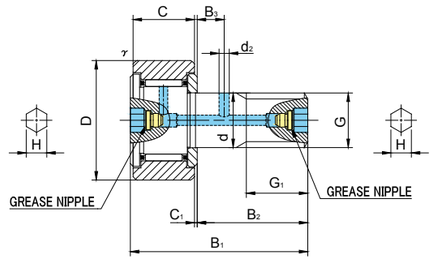


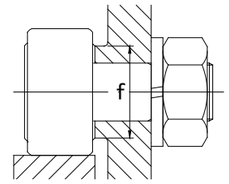


CF..AB



CF..VAB





CF..AB TYPE

Prepacked Grease

Stud diameter (mm)	h7 tolerance	Designation: Cylindrical outer ring, Without seals	Cylindrical outer ring, With seals	Crowned outer ring R1000(CF20 ≦), Without seals	Crowned outer ring R1000(CF20 ≦), With seals	D	C	d	G	G1	B1	B2	B3	C1	d2	H	r s min	f min	Basic dynamic load rating Cr N	Basic static load rating Cor N	Largest permissible load N	Limiting speed * rpm	Track load capacity: Cylindrical outer ring N	Track load capacity: Crowned outer ring N	Max tightening torque N·m	Mass g (approx)
20	0 -0.021	**CF20-1AB**	**CF20-1UUAB**	**CF20-1RAB**	**CF20-1UURAB**	**47**	24	20	M20×1.5	21	66	40.5	9	0.8	4	8	1	36	20 680	34 600	32 140	7 000	21 000	7 150	120	385
		CF20-1VAB	**CF20-1VUUAB**	**CF20-1VRAB**	**CF20-1VUURAB**														33 120	64 480	32 140	3 500				
24	0 -0.021	**CF24AB**	**CF24UUAB**	**CF24RAB**	**CF24UURAB**	**62**	29	24	M24×1.5	25	80	49.5	11	0.8	4	8	1	40	30 480	52 630	49 500	6 500	34 200	10 500	220	815
		CF24VAB	**CF24VUUAB**	**CF24VRAB**	**CF24VUURAB**														46 550	92 020	49 500	3 000				
24	0 -0.021	**CF24-1AB**	**CF24-1UUAB**	**CF24-1RAB**	**CF24-1UURAB**	**72**	29	24	M24×1.5	25	80	49.5	11	0.8	4	8	1	40	30 480	52 630	49 500	6 500	39 800	12 900	220	1 140
		CF24-1VAB	**CF24-1VUUAB**	**CF24-1VRAB**	**CF24-1VUURAB**														46 550	92 020	49 500	3 000				
30	0 -0.021	**CF30AB**	**CF30UUAB**	**CF30RAB**	**CF30UURAB**	**80**	35	30	M30×1.5	32	100	63	15	1	4	8	1	46	45 370	85 060	73 700	5 000	52 600	14 900	450	1 870
		CF30VAB	**CF30VUUAB**	**CF30VRAB**	**CF30VUURAB**														67 620	144 060	73 700	2 200				
30	0 -0.021	**CF30-1AB**	**CF30-1UUAB**	**CF30-1RAB**	**CF30-1UURAB**	**85**	35	30	M30×1.5	32	100	63	15	1	4	8	1	46	45 370	85 060	73 700	5 000	56 000	16 100	450	2 030
		CF30-1VAB	**CF30-1VUUAB**	**CF30-1VRAB**	**CF30-1VUURAB**														67 620	144 060	73 700	2 200				
30	0 -0.021	**CF30-2AB**	**CF30-2UUAB**	**CF30-2RAB**	**CF30-2UURAB**	**90**	35	30	M30×1.5	32	100	63	15	1	4	8	1	46	45 370	85 060	73 700	5 000	59 300	17 300	450	2 220
		CF30-2VAB	**CF30-2VUUAB**	**CF30-2VRAB**	**CF30-2VUURAB**														67 620	144 060	73 700	2 200				

CF..AB TYPE has oil holes (grease nipple) which are prepared in the stud head and thread side, also an oil hole is prepared in the outer surface of stud.

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

OUTER RINGS TOLERANCE (μm)

TYPE	Cylindrical outer ring	Crowned outer ring
CF10,CF10-1,CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

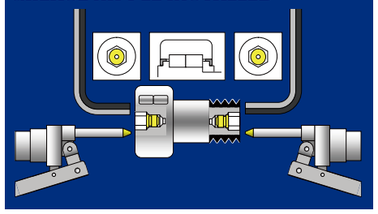
ACCESSORIES

TYPE	 Grease Nipple	 NUT
All types	Installed	Attached

CF..AB

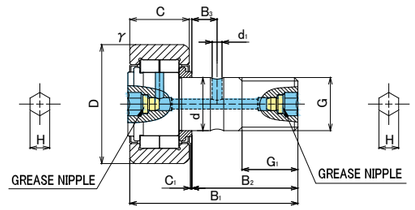
CAM FOLLOWERS

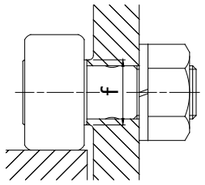
FULL COMPLEMENT DOUBLE ROW
HEXAGON SOCKET ON BOTH SIDES
GREASE NIPPLE INSTALLED





NUCF..AB





NUCF TYPE

Prepacked Grease

Stud diameter (mm)		Designation		Dimensions (mm)												Basic dynamic load rating	Basic static load rating	Largest permissible load	Limiting speed	Track load capacity		Max tightening torque	Mass
		Cylindrical outer ring	Crowned outer ring R500(NUCF16~NUCF18) R1000(NUCF20≦)																				
	h7 tolerance	Without seals	With seals	D	C	d	G	G1	B1	B2	B3	C1	D2	H	*r*s min	f	Cr N	Cor N	N	rpm	Cylindrical outer ring N	Crowned outer ring N	N·m	g (approx)
16	0 -0.018	**NUCF16AB**	**NUCF16RAB**	**35**	18	16	M16x1.5	17	52	32.5	7.8	0.8	3	6	0.6	20	23 400	27 200	11 900	5 200	11 200	3 140	58	167
18	0 -0.018	**NUCF18AB**	**NUCF18RAB**	**40**	20	18	M18x1.5	19	58	36.5	8	0.8	3	6	1	22	25 200	30 900	22 200	4 700	14 400	3 720	87	248
20	0 -0.021	**NUCF20AB**	**NUCF20RAB**	**52**	24	20	M20x1.5	21	66	40.5	9	0.8	4	8	1	31	43 000	58 100	25 700	3 300	23 200	8 230	120	461
	0 -0.021	**NUCF20-1AB**	**NUCF20-1RAB**	**47**	24	20	M20x1.5	21	66	40.5	9	0.8	4	8	1	27	38 900	48 900	25 300	3 800	21 000	7 150	120	390
24	0 -0.021	**NUCF24AB**	**NUCF24RAB**	**62**	28	24	M24x1.5	25	80	49.5	11	1.3	4	8	1	38	57 600	74 300	35 000	2 800	32 000	10 500	220	789
	0 -0.021	**NUCF24-1AB**	**NUCF24-1RAB**	**72**	28	24	M24x1.5	25	80	49.5	11	1.3	4	8	1.1	44	63 300	87 500	53 700	2 300	37 200	12 900	220	1 020
30	0 -0.021	**NUCF30AB**	**NUCF30RAB**	**80**	35	30	M30x1.5	32	100	63	15	1	4	8	1.1	47	94 800	135 700	73 700	2 100	52 600	14 900	450	1 630
	0 -0.021	**NUCF30-2AB**	**NUCF30-2RAB**	**90**	35	30	M30x1.5	32	100	63	15	1	4	8	1.1	47	94 800	135 700	73 700	2 100	59 300	17 300	450	1 990

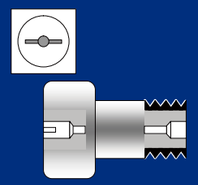
OUTER RINGS TOLERANCE (μm)

TYPE	Cylindrical outer ring	Crowned outer ring
NUCF16,NUCF18,NUCF20-1	0/-11	0/-50
NUCF20,NUCF24,NUCF24-1,NUCF30	0/-13	0/-50
NUCF30-2	0/-15	0/-50

ACCESSORIES

TYPE	 Grease Nipple	 NUT
All types	Installed	Attached

CAM FOLLOWERS
SCREWDRIVER SLOT HEAD

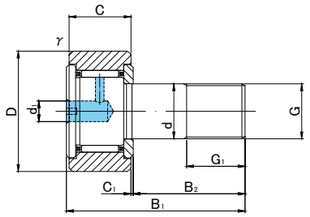




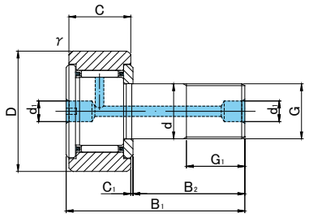
CF



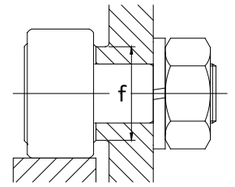
CF..V



CF5(V)～CF10−1(V)



CF10(V)K～CF10−1(V)K



CF TYPE

Prepacked Grease

Stud diameter (mm)	h7 tolerance	Designation: Cylindrical outer ring, Without seals	Designation: Cylindrical outer ring, With seals	Designation: Crowned outer ring R250(CF5) R500(CF6 ～ CF10-1), Without seals	Designation: Crowned outer ring R250(CF5) R500(CF6 ～ CF10-1), With seals	D	C	d	G	G1	B1	B2	B3	C1	d1	d2	r s min	f min	Basic dynamic load rating Cr N	Basic static load rating Cor N	Largest permissible load N	Limiting speed * rpm	Track load capacity: Cylindrical outer ring N	Track load capacity: Crowned outer ring N	Max tightening torque N·m	Mass g (approx)
5	0 -0.012	CF 5	CF 5UU	CF 5R	CF 5UUR	13	9	5	M5×0.8	7.5	23	13	—	0.5	3.1	—	0.3	9.7	3 140	2 770	1 420	29 000	2 250	780	2	10.5
		CF 5V	CF 5VUU	CF 5VR	CF 5VUUR														5 100	5 500	1 420	11 600				
6	0 -0.012	CF 6	CF 6UU	CF 6R	CF 6UUR	16	11	6	M6×1	8	28	16	—	0.6	4	—	0.3	11	3 630	3 630	2 110	25 000	3 430	1 080	3	18.5
		CF 6V	CF 6VUU	CF 6VR	CF 6VUUR														6 960	8 530	2 110	12 000				
8	0 -0.015	CF 8	CF 8UU	CF 8R	CF 8UUR	19	11	8	M8×1.25	10	32	20	—	0.6	4	—	0.3	13	4 310	4 710	4 710	20 000	4 020	1 370	8	28.5
		CF 8V	CF 8VUU	CF 8VR	CF 8VUUR														8 130	11 170	4 710	9 000				
10	0 -0.015	CF10	CF10UU	CF10R	CF10UUR	22	12	10	M10×1.25	12	36	23	—	0.6	4	—	0.3	15	5 390	6 860	6 860	17 000	4 700	1 670	15	45
		CF10V	CF10VUU	CF10VR	CF10VUUR														9 510	14 500	7 450	7 500				
10	0 -0.015	CF10-1	CF10-1UU	CF10-1R	CF10-1UUR	26	12	10	M10×1.25	12	36	23	—	0.6	4	—	0.3	15	5 390	6 860	6 860	17 000	5 490	2 060	15	60
		CF10-1V	CF10-1VUU	CF10-1VR	CF10-1VUUR														9 510	14 500	7 450	7 500				
10	0 -0.015	CF10K	CF10UUK	CF10RK	CF10UURK	22	12	10	M10×1	12	36	23	—	0.6	4	—	0.3	15	5 390	6 860	6 860	17 000	4 700	1 670	15	45
		CF10VK	CF10VUUK	CF10VRK	CF10VUURK														9 510	14 500	7 450	7 500				
10	0 -0.015	CF10-1K	CF10-1UUK	CF10-1RK	CF10-1UURK	26	12	10	M10×1	12	36	23	—	0.6	4	—	0.3	15	5 390	6 860	6 860	17 000	5 490	2 060	15	60
		CF10-1VK	CF10-1VUUK	CF10-1VRK	CF10-1VUURK														9 510	14 500	7 450	7 500				

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

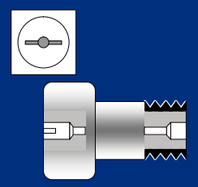
OUTER RINGS TOLERANCE

TYPE	Cylindrical outer ring	Crowned outer ring
CF5,CF6	0/-8	0/-50
CF8,CF10,CF10-1,CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	 STOP PLUG	 RESIN PLUG	 NUT
CF5	—	—	Attached
CF6/CF8/CF10/CF10-1	—	ϕ 4 Attached	Attached
CF12/CF12-1/CF16/CF18	ϕ 6 Attached	ϕ 6 Attached	Attached
CF20/CF20-1/CF24/CF24-1/CF30/CF30-1/CF30-2	ϕ 8 Attached	ϕ 8 Attached	Attached

CAM FOLLOWERS
SCREWDRIVER SLOT HEAD

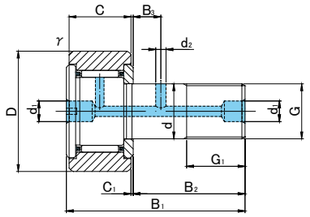


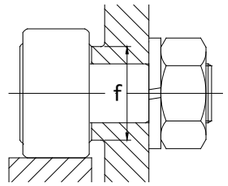


CF



CF..V





CF TYPE

Prepacked Grease

Stud diameter (mm)		Designation: Cylindrical outer ring		Designation: Crowned outer ring R500(CF12 ～ CF18) R1000(CF20 ≦)		Dimensions (mm)												Basic dynamic load rating	Basic static load rating	Largest permissible load	Limiting speed *	Track load capacity		Max tightening torque	Mass
	h7 tolerance	Without seals	With seals	Without seals	With seals	D	C	d	G	G1	B1	B2	B3	C1	d1	d2	r s min	f min	Cr N	Cor N	N	rpm	Cylindrical outer ring N	Crowned outer ring N	N·m	g (approx)
12	0 -0.018	**CF12**	**CF12UU**	**CF12R**	**CF12UUR**	**30**	14	12	M12×1.5	13	40	25	6	0.6	6	3	0.6	20	7 940	9 800	9 800	14 000	7 060	2 450	22	95
		CF12V	**CF12VUU**	**CF12VR**	**CF12VUUR**														13 430	19 700	11 270	6 000				
12	0 -0.018	**CF12-1**	**CF12-1UU**	**CF12-1R**	**CF12-1UUR**	**32**	14	12	M12×1.5	13	40	25	6	0.6	6	3	0.6	20	7 940	9 800	9 800	14 000	7 450	2 740	22	105
		CF12-1V	**CF12-1VUU**	**CF12-1VR**	**CF12-1VUUR**														13 430	19 700	11 270	6 000				
16	0 -0.018	**CF16**	**CF16UU**	**CF16R**	**CF16UUR**	**35**	18	16	M16×1.5	17	52	32.5	8	0.8	6	3	0.6	24	12 050	18 330	18 330	10 000	11 200	3 140	58	170
		CF16V	**CF16VUU**	**CF16VR**	**CF16VUUR**														20 680	37 630	19 800	4 500				
18	0 -0.018	**CF18**	**CF18UU**	**CF18R**	**CF18UUR**	**40**	20	18	M18×1.5	19	58	36.5	8	0.8	6	3	1	26	14 700	25 200	25 200	8 500	14 400	3 720	87	250
		CF18V	**CF18VUU**	**CF18VR**	**CF18VUUR**														25 280	51 350	26 560	3 500				
20	0 -0.021	**CF20**	**CF20UU**	**CF20R**	**CF20UUR**	**52**	24	20	M20×1.5	21	66	40.5	9	0.8	8	4	1	36	20 680	34 600	32 140	7 000	23 200	8 230	120	460
		CF20V	**CF20VUU**	**CF20VR**	**CF20VUUR**														33 120	64 480	32 140	3 500				
20	0 -0.021	**CF20-1**	**CF20-1UU**	**CF20-1R**	**CF20-1UUR**	**47**	24	20	M20×1.5	21	66	40.5	9	0.8	8	4	1	36	20 680	34 600	32 140	7 000	21 000	7 150	120	385
		CF20-1V	**CF20-1VUU**	**CF20-1VR**	**CF20-1VUUR**														33 120	64 480	32 140	3 500				
24	0 -0.021	**CF24**	**CF24UU**	**CF24R**	**CF24UUR**	**62**	29	24	M24×1.5	25	80	49.5	11	0.8	8	4	1	40	30 480	52 630	49 500	6 500	34 200	10 500	220	815
		CF24V	**CF24VUU**	**CF24VR**	**CF24VUUR**														46 550	92 020	49 500	3 000				
24	0 -0.021	**CF24-1**	**CF24-1UU**	**CF24-1R**	**CF24-1UUR**	**72**	29	24	M24×1.5	25	80	49.5	11	0.8	8	4	1	40	30 480	52 630	49 500	6 500	39 800	12 900	220	1 140
		CF24-1V	**CF24-1VUU**	**CF24-1VR**	**CF24-1VUUR**														46 550	92 020	49 500	3 000				
30	0 -0.021	**CF30**	**CF30UU**	**CF30R**	**CF30UUR**	**80**	35	30	M30×1.5	32	100	63	15	1	8	4	1	46	45 370	85 060	73 700	5 000	52 600	14 900	450	1 870
		CF30V	**CF30VUU**	**CF30VR**	**CF30VUUR**														67 620	144 060	73 700	2 200				
30	0 -0.021	**CF30-1**	**CF30-1UU**	**CF30-1R**	**CF30-1UUR**	**85**	35	30	M30×1.5	32	100	63	15	1	8	4	1	46	45 370	85 060	73 700	5 000	56 000	16 100	450	2 030
		CF30-1V	**CF30-1VUU**	**CF30-1VR**	**CF30-1VUUR**														67 620	144 060	73 700	2 200				
30	0 -0.021	**CF30-2**	**CF30-2UU**	**CF30-2R**	**CF30-2UUR**	**90**	35	30	M30×1.5	32	100	63	15	1	8	4	1	46	45 370	85 060	73 700	5 000	59 300	17 300	450	2 220
		CF30-2V	**CF30-2VUU**	**CF30-2VR**	**CF30-2VUUR**														67 620	144 060	73 700	2 200				

* Without seals, suitable for grease lubrication. In case of oil lubrication, up to 130% of this value shall be permissible, and 70% of this value shall apply for types with seals.

OUTER RINGS TOLERANCE

TYPE	Cylindrical outer ring	Crowned outer ring
CF5,CF6	0/-8	0/-50
CF8,CF10,CF10-1,CF12	0/-9	0/-50
CF12-1,CF16,CF18,CF20-1	0/-11	0/-50
CF20,CF24,CF24-1,CF30	0/-13	0/-50
CF30-1,CF30-2	0/-15	0/-50

ACCESSORIES

TYPE	 STOP PLUG	 RESIN PLUG	 NUT
CF5	—	—	Attached
CF6/CF8/CF10/CF10-1	—	ϕ 4 Attached	Attached
CF12/CF12-1/CF16/CF18	ϕ 6 Attached	ϕ 6 Attached	Attached
CF20/CF20-1/CF24/CF24-1/CF30/CF30-1/CF30-2	ϕ 8 Attached	ϕ 8 Attached	Attached